

# Sensitivity Analysis for Nonlinear Models with a Synthetic Confounder.

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## Abstract

This paper propose a sensitivity analysis approach for general likelihood-based models. A synthetic confounder is constructed via simulation which has pre-specified partial correlations with the residualised treatment and outcome variables, thus capturing researchers' beliefs about the strength of the confounding. By including the synthetic confounder in the short (misspecified) model, the framework yields adjusted average marginal effects. The paper shows how to calculate fragility thresholds via a bisection algorithm and includes a benchmarking procedure to calibrate the sensitivity parameters against observed covariates. A Monte Carlo simulation shows that the new framework outperforms direct application of existing sensitivity approaches for linear models under moderate to strong confounding. The new framework is applied the 1970 British Cohort Study to evaluate the sensitivity of the effect of early life cognitive skills on labour market participation and job quality during adulthood.

**Key Words:** *Sensitivity analysis, omitted variable bias, nonlinear models, average marginal effects, synthetic confounder, fragility thresholds*

**JEL Classification:** *C25, C35, C52, C18.*

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# 1 Introduction

The omission of relevant variables from models is one of the main concerns in empirical economics and statistics. Excluding a variable that is correlated with both the outcome and a predictor of interest (e.g. a treatment indicator) causes the estimated coefficient of the predictor to absorb part of the omitted variable’s influence, producing a bias. In particular, the resulting estimates cannot be given a causal interpretation because they confound the effect of the predictor with the effect of the omitted variable or confounder, and policy conclusions drawn from them may be misleading.

Starting with the seminal contributions of Cornfield et al. (1959); Bross (1966, 1967) and Rosenbaum and Rubin (1983), numerous methods have been developed to assess how sensitive the conclusions of observational studies are to the possible presence of such confounders (Altonji et al., 2005; Oster, 2019; VanderWeele and Arah, 2011; Ding and VanderWeele, 2016; Carnegie et al., 2016; Blackwell, 2013; Frank, 2000; Cinelli and Hazlett, 2020; Masten and Poirier, 2022; Sjölander et al., 2026; Brumback et al., 2004, among others). These methods explore how strong would an omitted confounder need to be to overturn the result of a study. If moderate confounding is sufficient to overturn the findings, then the conclusions of the study are fragile and should be treated with caution regardless of their statistical significance.

A large proportion of the results in the sensitivity literature have focused on linear models. Not only do they constitute the workhorse of modern empirical practice, but they also often allow for closed form sensitivity formulas with known properties (e.g. Oster, 2019; Cinelli and Hazlett, 2020). However, researchers often need to deal with discrete outcomes or outcomes whose analysis benefit from the use of nonlinear models for precision and efficiency reasons. The question is whether existing linear methods can be applied to these nonlinear models (effectively using a linear probability approximation) and, if not, which alternative approaches can be considered<sup>1</sup>.

This paper explores both questions by studying how much Average Marginal Effects

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<sup>1</sup>This is a general question which does not depend on whether the researcher is willing to maintain a selection-on-observables assumption (with the intention of establishing a causal claim).

(AME) from a likelihood-based nonlinear model would change if there were an omitted variable with given partial correlations with the treatment and outcome residuals. The paper makes three principal contributions. First, it documents the omitted variable bias of general likelihood-based models, showing that it admits a Frisch-Waugh-Lovell interpretation as the slope coefficient of a linear regression. This suggests that the framework in Cinelli and Hazlett (2020) could be directly applied to nonlinear models replacing OLS residuals with, for example, standardised residuals. However, the bias includes a remainder term that reflects the curvature of the log-likelihood and, more importantly, grows quadratically with the strength of the confounder. As a result, the direct application of linear methods in nonlinear settings is valid as first-order approximations but only when confounding is weak. Therefore, there is scope for the development of bespoke methods for nonlinear models.

The paper proposes a new simulation based approach to sensitivity analysis in nonlinear models. The method is inspired by Cinelli and Hazlett (2020), who derive exact expressions for the omitted variable bias in linear models as a function of the partial correlations of the omitted variable with the residualised treatment and outcome. In nonlinear models, confounders with the same two partial correlations can generate different biases. The problem thus reduces to constructing a tractable member of the class of possible confounders with specific correlations with the residualised treatment and outcome. Because covariance is a linear operator, we can then build the synthetic confounder as a linear combination of the residualised treatment and outcome, ensuring the desired correlations. By augmenting the short (misspecified) model with this synthetic confounder and re-estimating by maximum likelihood, the method delivers the adjusted AME, which is the starting point for a full sensitivity analysis. This framework will apply to likelihood-based models, including logit, probit, complementary log-log, Poisson, negative binomial, ordered logit, and multinomial logit, among others.

The sensitivity parameters in the synthetic confounder approach are partial correlations between the synthetic confounder and the residualised treatment and outcome ( $\rho_Y$  and  $\rho_D$  respectively). Therefore, they are bounded in  $[-1, 1]$ , simple to interpret, and

directly comparable across models and datasets. Following Cinelli and Hazlett (2020) and Masten and Poirier (2022), the framework enables the calculation of explain-away and sign-change thresholds for every value of  $\rho_Y$  in the grid. This is done through a grid-search plus bisection algorithm. As in Cinelli and Hazlett (2020), the paper develops a benchmarking procedure that calibrates the sensitivity parameters against observed covariates, grounding the analysis in the specific empirical setting.

The new method circumvents, to a great extent, the main simplifying assumptions identified by Ding and VanderWeele (2016). Specifically, while the discussion assumes homogeneity<sup>2</sup> for expositional convenience, this assumption is not required for the use of the method. The method performs sensitivity analysis under the full range of alternative hypotheses rather than only under the null of no effect by delivering an adjusted AME at every feasible  $(\rho_D, \rho_Y)$  point in the sensitivity parameter space. Furthermore, the asymptote problem identified by Masten and Poirier (2022) for the  $\delta$  parameter in Oster (2019) does not arise in our framework because  $(\rho_D, \rho_Y)$  are bounded partial correlations lying in a compact set and therefore large sign-change thresholds are possible without the denominator collapse that causes Oster’s  $\delta$  to diverge.

The paper illustrates the application of the method in an evaluation of the sensitivity of cognitive skills as a predictor of labour market participation and job quality. Using data from the 1970 British Cohort Study (BCS70), we estimate short models of full-time employment and managerial or supervisory status at age 46 as a function of a composite measure of cognitive ability assessed at age 10, controlling for a rich set of early-life covariates including perinatal characteristics, parental background, and teacher-reported measures of developmental behaviour. Because cognitive skills are measured before the adult outcomes of interest are realised, the design reduces concerns about reverse causality. However, unobserved variables that are associated with both childhood cognitive skill and adult labour-market success (such as family socioeconomic advantage or parental ability) would distort the short model’s estimates of the causal effect of cognition. The synthetic confounder method allows us to explore how strongly correlated

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<sup>2</sup>No interaction between the confounder and the treatment.

with both cognitive skill and labour-market outcomes would an omitted variable need to be before the conclusions change.

This is not the first paper to propose sensitivity methods for nonlinear models. Indeed, by working within a potential outcomes framework, the literature has left distributional assumptions implicit. For example, Imbens (2003); Brumback et al. (2004); Blackwell (2013) work within a potential outcomes framework that imposes no distributional assumption on the outcome at the identification level. However, Imbens requires a normal linear structure to derive closed-form sensitivity contours after integrating out the unobserved confounder, while the outcome adjustment in Blackwell is additive and its  $R^2$  reparameterisation relies on a linear variance decomposition, neither of which extends naturally to likelihood-based nonlinear models. Brumback et al. (2004) develop a sensitivity analysis framework designed for time-varying treatments estimated by inverse probability weighting and does not extend naturally to the likelihood-based cross-sectional models. In particular, their framework does not deliver adjusted average marginal effects at each sensitivity parameter value.

VanderWeele and Arah (2011) derive bounds for the bias due to unmeasured confounding for general outcomes within a potential outcomes framework, under a homogeneity assumption. Ding and VanderWeele (2016) introduce the E-value, (the minimum strength of association that an unmeasured confounder would need to have with both treatment and outcome to fully explain away an observed association) within a binary outcomes and exposures framework. Sjölander et al. (2026) extend the E-value to marginal causal effects, however their framework retains binary outcomes and exposures and it produces bounds on the causal effect rather than adjusted point estimates.

The paper closest to ours is Carnegie et al. (2016), who introduce a simulated approach in which a synthetic confounder is drawn from the conditional distribution implied by a specified complete-data likelihood, for pre-specified values of the confounder's coefficient in the treatment and outcome models. When both the outcome and treatment are continuous and normally distributed, this conditional distribution is available in closed form. Park et al. (2026) extend the simulation-based sensitivity analysis of Carnegie et al. (2016)

to causal decomposition analysis with optimal treatment regimes, and develop a formal benchmarking strategy for binary treatments using original data scales. The method presented here differs from theirs in four respects. First, the method is parameterised in terms of partial correlations rather than coefficients, making the sensitivity parameters scale-free and directly comparable across models. Second, the approach defines a closed-form feasibility constraint on achievable parameter pairs. The method supports the full range of models used in applied economics, including Poisson, negative binomial, ordered logit, and multinomial logit. Finally, a bisection algorithm is implemented to compute fragility thresholds analytically rather than reading them from a contour plot.

The remainder of the paper proceeds as follows. Section 2 derives the approximate bias formula for likelihood-based models, revealing how second-order curvature terms invalidate linear approximations under strong confounding. Section 3 develops the simulation-based sensitivity analysis framework. Section 4 introduces fragility thresholds and benchmarking procedures that allow researchers to calibrate sensitivity parameters against observed covariates. Section 5 presents a Monte Carlo simulation to evaluate the method. Section 6 contains the empirical application, while section 7 concludes. Proofs of the main results and additional supporting material are deferred to an Appendix.

## 2 Omitted Variable Bias.

Assume the existence of a sample of  $i = 1, \dots, n$  observations. Let  $Y_i$  denote an outcome variable with range depending on the model, (e.g.  $Y_i \in \{0, 1\}$  for binary models,  $Y_i \in [0, 1]$  for fractional models,  $Y_i \in \{1, \dots, J\}$  for multinomial models, etc). Let  $D_i \in \mathbb{R}$  be the variable of interest (e.g. a treatment indicator or a continuous measure of income or cognitive skills) and  $\mathbf{X}_i \in \mathbb{R}^p$  a vector of observed covariates. In what follows, to fix ideas, we give  $D$  the interpretation of a treatment variable in the causal inference sense. The variable  $Z_i \in \mathbb{R}$  is an unobserved confounder. Define  $A_i = (1, \mathbf{X}_i)'$ ,  $W_i^* = (A_i', D_i)'$ , and  $W_i = (A_i', D_i, Z_i)'$ . Thus,  $A_i$  contains the intercept and the observed covariates  $\mathbf{X}_i$ , and  $\boldsymbol{\alpha} \in \mathbb{R}^{p+1}$  denotes the corresponding nuisance parameter vector, including the

intercept. Let  $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \beta)' \in \mathbb{R}^{p+2}$  denote the short-model parameter vector, and let  $\boldsymbol{\xi} = (\boldsymbol{\theta}', \delta)' \in \mathbb{R}^{p+3}$  denote the corresponding long-model parameter vector. The parameter  $\beta$  is associated with  $D_i$ , while  $\delta$  is associated with  $Z_i$ . Let  $f(y, w; \boldsymbol{\xi})$  denote the joint model density or mass function for  $(Y_i, W_i)$  under the long model. The log-likelihood, score, and Hessian contributions of observation  $i$  under the full (long) model are

$$\ell_i(\boldsymbol{\xi}) = \log f(Y_i, W_i; \boldsymbol{\xi}), \quad \mathbf{s}_i(\boldsymbol{\xi}) = \frac{\partial \ell_i(\boldsymbol{\xi})}{\partial \boldsymbol{\xi}}, \quad \mathbf{H}_i(\boldsymbol{\xi}) = \frac{\partial^2 \ell_i(\boldsymbol{\xi})}{\partial \boldsymbol{\xi} \partial \boldsymbol{\xi}'}. \quad (2.1)$$

Let  $H_{i,\alpha\alpha}(\boldsymbol{\xi})$  denote the nuisance block of the long-model Hessian corresponding to  $\boldsymbol{\alpha}$ , and let  $H_{i,\beta\delta}(\boldsymbol{\xi})$ ,  $H_{i,\beta\beta}(\boldsymbol{\xi})$ ,  $H_{i,\beta\alpha}(\boldsymbol{\xi})$ , and  $H_{i,\alpha\delta}(\boldsymbol{\xi})$  denote the corresponding blocks of  $\mathbf{H}_i(\boldsymbol{\xi})$  accordingly. The population score and information matrix are

$$\mathbf{s}(\boldsymbol{\xi}) := \mathbb{E}[\mathbf{s}_i(\boldsymbol{\xi})], \quad \mathcal{I}(\boldsymbol{\xi}) := -\mathbb{E}[\mathbf{H}_i(\boldsymbol{\xi})], \quad (2.2)$$

where the expectation is taken under the true data-generating process. Because  $Z_i$  is unobservable, the researcher fits the short model that omits  $Z_i$  (implicitly setting  $\delta = 0$ ). The short-model log-likelihood, score, and Hessian are

$$\ell_i^*(\boldsymbol{\theta}) = \ell_i(\boldsymbol{\theta}, 0), \quad \mathbf{s}_i^*(\boldsymbol{\theta}) = \frac{\partial \ell_i^*(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}, \quad \mathbf{H}_i^*(\boldsymbol{\theta}) = \frac{\partial^2 \ell_i^*(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}. \quad (2.3)$$

The short-model maximum likelihood estimator  $\hat{\boldsymbol{\theta}}^*$  solves  $n^{-1} \sum_i \mathbf{s}_i^*(\hat{\boldsymbol{\theta}}^*) = \mathbf{0}$  and converges in probability<sup>3</sup> to the pseudo-true parameter  $\boldsymbol{\theta}^*$ , defined as:

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} \mathbb{E}[\ell_i^*(\boldsymbol{\theta})], \quad (2.4)$$

which is equivalently the minimiser of the Kullback–Leibler divergence from the true density to the short-model density (White, 1982). Because the short model omits a relevant variable, the pseudo-true and true parameter values do not coincide, in general. The magnitude of the bias is established in the following results (the proofs and specific underpinning assumptions are given in the Appendix A ).

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<sup>3</sup>Under regularity conditions detailed in Appendix A.

**Lemma 2.1** (Nonlinear OVB). *Under regularity conditions **R1–R7** (detailed in Appendix A), the pseudo-true parameter satisfies*

$$\boldsymbol{\theta}^* - \boldsymbol{\theta}_0 = \boldsymbol{\Pi} \delta_0 + O(\delta_0^2), \quad (2.5)$$

where

$$\boldsymbol{\Pi} = [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \mathcal{I}_{\boldsymbol{\theta}\delta}(\boldsymbol{\xi}_0) \in \mathbb{R}^{p+2}, \quad (2.6)$$

$\mathcal{I}^*(\boldsymbol{\theta}_0) = -\mathbb{E}[\mathbf{H}_i^*(\boldsymbol{\theta}_0)]$  is the short-model information matrix,  $\mathcal{I}_{\boldsymbol{\theta}\delta}(\boldsymbol{\xi}_0) = -\mathbb{E}[H_{i,\boldsymbol{\theta}\delta}(\boldsymbol{\xi}_0)]$  is the  $\boldsymbol{\theta}$ – $\delta$  cross-block of the long-model information matrix, and  $\boldsymbol{\xi}_0 = (\boldsymbol{\theta}'_0, 0)'$ . In particular,  $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\| = O(\delta_0)$  as  $\delta_0 \rightarrow 0$ .

As shown in Appendix A.3, the quantity  $\boldsymbol{\Pi}$  can be expressed as

$$\boldsymbol{\Pi} = [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \text{Cov}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)], \quad (2.7)$$

where  $\mathbf{s}_i^*(\boldsymbol{\theta}_0)$  is the short-model score at the truth and  $s_{i,\delta}(\boldsymbol{\theta}_0, 0) = \partial \ell_i / \partial \delta|_{\delta_0=0}$  is the score of the true model with respect to the omitted coefficient evaluated at the correctly specified parameter. This covariance is zero if and only if  $Z_i$  carries no information about the short-model residuals at the truth. In that case omitting  $Z_i$  causes no first-order distortion of the pseudo-true parameter.<sup>4</sup> The next results specialises to the treatment parameter.

**Corollary 2.1** (Treatment coefficient bias). *Under the conditions of Lemma A.2, the bias in the treatment coefficient is*

$$\beta^* - \beta_0 = \tilde{\pi} \delta_0 + O(\delta_0^2), \quad (2.8)$$

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<sup>4</sup>The remainder is zero in the linear case. Linear models solve a problem of the form  $\ell_i(\boldsymbol{\xi}) = -\frac{1}{2\sigma^2}(Y_i - W_i'\boldsymbol{\xi})^2 + \text{const}$ , which is quadratic in  $\boldsymbol{\xi}$ . Its Hessian  $\mathbf{H}_i(\boldsymbol{\xi}) = -W_i W_i' / \sigma^2$  is therefore constant. As a result, the Taylor expansion of the population score around  $\boldsymbol{\theta}^*$  is exact at second order, and the implicit-function expansion of  $\boldsymbol{\theta}^*(\delta_0)$  around  $\delta_0 = 0$  (Lemma A.2) terminates exactly at first order. The remainder is therefore identically zero, and the main results in Cinelli and Hazlett (2020) are recovered as a special case of Corollary A.1.

where

$$\tilde{\pi} = \frac{\tilde{\mathcal{C}}_{\beta\delta}^0}{\tilde{\mathcal{I}}_{\beta\beta}^0} = \frac{\text{Cov}_0(\tilde{s}_{\beta,i}, \tilde{s}_{\delta,i})}{\text{Var}_0(\tilde{s}_{\beta,i})}, \quad (2.9)$$

and  $\tilde{s}_{\beta,i}$  and  $\tilde{s}_{\delta,i}$  are the residualised scores obtained by projecting the treatment score  $s_{\beta,i}^*$  and the confounding score  $s_{i,\delta}$  onto the nuisance score  $\mathbf{s}_{\alpha,i}^*$ , all evaluated at  $\boldsymbol{\xi}_0 = (\boldsymbol{\theta}'_0, 0)'$ . Here  $s_{\beta,i}^*$  and  $\mathbf{s}_{\alpha,i}^*$  are the  $\beta$ - and  $\boldsymbol{\alpha}$ -components of the short-model score  $\mathbf{s}_i^*(\boldsymbol{\theta}_0)$ , and  $s_{i,\delta} = s_{i,\delta}(\boldsymbol{\theta}_0, 0)$  is the  $\delta$ -component of the long-model score.

The leading term of the bias depends on  $Z$  only through how it relates to  $Y$  (captured by  $\delta_0$ ) and  $D$  (via  $\tilde{\pi}$ ). In particular  $\tilde{\pi}$  measures the strength of the association between the treatment and the confounder after partialling out the nuisance covariates. The bias is the product of these two terms. As a result, a confounder that strongly affects the outcome ( $\delta_0$  large) causes negligible bias if it is orthogonal to treatment ( $\tilde{\pi}$  close to 0). Similarly, a confounder strongly associated with treatment would cause negligible bias if it does not affect the outcome.

As discussed in the Appendix (proof of Corolary A.1 and section A.4) the residualised-score representation in (2.9) gives  $\tilde{\pi}$  a Frisch–Waugh–Lovell interpretation. The bias in  $\beta$  is determined by the association between the part of the treatment score unexplained by the nuisance scores and the part of the confounding score unexplained by those same nuisance scores. This interpretation is clearer for GLMs with a canonical link function, given the structure of the Hessian. As shown in Appendix A.4, for GLMs with canonical link and variance weight  $w_i^* = V(\mu_i(\boldsymbol{\xi}^*))$  evaluated at  $\boldsymbol{\xi}^* = (\boldsymbol{\theta}^{*'}, 0)'$ ,

$$\tilde{\mathcal{I}}_{\beta\beta}^* = \mathbb{E}[w_i^* \tilde{D}_i^{*2}], \quad \tilde{\mathcal{I}}_{\beta\delta}^* = \mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*], \quad (2.10)$$

where

$$\begin{aligned} \tilde{D}_i^* &= D_i - (\boldsymbol{\pi}_D^*)' A_i, & \boldsymbol{\pi}_D^* &= \mathbb{E}[w_i^* A_i A_i']^{-1} \mathbb{E}[w_i^* A_i D_i], \\ \tilde{Z}_i^* &= Z_i - (\boldsymbol{\pi}_Z^*)' A_i, & \boldsymbol{\pi}_Z^* &= \mathbb{E}[w_i^* A_i A_i']^{-1} \mathbb{E}[w_i^* A_i Z_i], \end{aligned}$$

are residuals from  $w^*$ -weighted projections of  $D_i$  and  $Z_i$  onto  $A_i = (1, \mathbf{X}_i')' \in \mathbb{R}^{p+1}$ , and

so

$$\tilde{\pi} = \frac{\mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*]}{\mathbb{E}[w_i^* \tilde{D}_i^{*2}]}.$$
 (2.11)

This is a second-moment ratio and, under the same additional differentiability assumption,  $w_i^* = w_i^0 + O(\delta_0)$  where  $w_i^0 = V(\mu_i(\boldsymbol{\xi}_0))$ , and (2.11) becomes

$$\tilde{\pi} = \frac{\text{Cov}_{w^0}(\tilde{D}_i^0, \tilde{Z}_i^0)}{\text{Var}_{w^0}(\tilde{D}_i^0)} + O(\delta_0),$$
 (2.12)

where  $\tilde{D}_i^0$  and  $\tilde{Z}_i^0$  are the residuals from  $w^0$ -weighted projections of  $D_i$  and  $Z_i$  onto  $A_i$ , and  $\text{Cov}_{w^0}$ ,  $\text{Var}_{w^0}$  denote  $w^0$ -weighted covariance and variance. Hence, in a neighbourhood of  $\delta_0 = 0$ ,  $\tilde{\pi}$  is approximately the population regression slope from regressing  $\tilde{Z}_i^0$  on  $\tilde{D}_i^0$  with weights  $w_i^0$ .

For linear models, Cinelli and Hazlett (2020) show that

$$\beta^* - \beta_0 = \delta_0 \underbrace{\frac{\text{Cov}(\dot{Z}_i, \dot{D}_i)}{\text{Var}(\dot{D}_i)}}_{\tilde{\pi}}$$
 (2.13)

exactly, where  $\dot{D}_i = D_i^\perp \mathbf{X}$  and  $\dot{Z}_i = Z_i^\perp \mathbf{X}$  are the residuals from OLS projections of  $D$  and  $Z$  onto  $A = (1, \mathbf{X}')$ . The structural similarity of (2.12) with (2.13) might suggest that the results for linear models in Cinelli–Hazlett could be applied to likelihood-based models using a suitably defined residual. For example, one could estimate a linear probability model and apply Cinelli–Hazlett directly. However, by Lemma 2.1 this procedure is likely to work only when the amount of confounding is very small. The nonlinear OVB formula contains a non-trivial remainder of order  $O(\delta_0^2)$ . This implies that stronger confounding produces a larger remainder and a more severe approximation error.

### 3 Synthetic Confounder Sensitivity Analysis

The remainder in the nonlinear OVB formula prevents exact closed-form characterisation of the bias. However, closed-form bounds are not essential to develop a viable sensitivity framework for nonlinear models. As in the linear case, it suffices to characterise how

the confounder relates to the outcome and the treatment after removing the effect of the ancillary regressors. Specifically, Cinelli and Hazlett (2020) show that (2.13) may be written as

$$\beta^* - \beta_0 = \text{sign}(\beta^*) \sqrt{\frac{R_{Y \sim Z|D, \mathbf{X}}^2 R_{D \sim Z|\mathbf{X}}^2}{1 - R_{D \sim Z|\mathbf{X}}^2}} \frac{\text{sd}(\dot{Y})}{\text{sd}(\dot{D})}. \quad (3.1)$$

The partial  $R^2$  values are defined in terms of residuals

$$R_{Y \sim Z|\mathbf{X}, D}^2 = \text{cor}^2(\dot{Y}_i, \dot{Z}_{Y,i}), \quad R_{D \sim Z|\mathbf{X}}^2 = \text{cor}^2(\dot{D}_i, \dot{Z}_{D,i}), \quad (3.2)$$

where  $\dot{Y}_i = Y_i^{\perp D, \mathbf{X}}$  and  $\dot{D}_i = D_i^{\perp \mathbf{X}}$  denote the OLS residuals from regressing  $Y$  on  $(D, \mathbf{X})$  and  $D$  on  $\mathbf{X}$ , respectively and  $\dot{Z}_{Y,i} = Z_i^{\perp D, \mathbf{X}}$  and  $\dot{Z}_{D,i} = Z_i^{\perp \mathbf{X}}$  are the confounder residualised on the same variables as the outcome and the treatment.

The two quantities  $R_{Y \sim Z|\mathbf{X}, D}^2$  and  $R_{D \sim Z|\mathbf{X}}^2$  therefore characterise how strongly the confounder is related to the residual variation in  $Y$  and  $D$  that remains after conditioning on the relevant observed variables. They do not uniquely determine  $Z$ , but they define a set of possible confounders that all have these two partial correlations. In a linear model, this distinction does not matter. The OVB formula depends on the confounder only through these two partial correlations, so every confounder satisfying these restrictions generates the same bias. Hence, the two  $R^2$  measures are sufficient to determine the bias without specifying any further features of  $Z$ .

In a nonlinear model, however, confounders with the same two partial correlations can generate different biases because the nonlinear OVB expression contains higher-order terms that depend on features of the confounder beyond these correlations. To progress beyond this limitation, we then need to construct one analytically tractable member of the set of possible confounders. The natural choice is to define  $Z$  as a linear combination of the treatment and outcome residuals, with the coefficients chosen to deliver the specified partial correlations exactly. This provides a closed-form representative confounder and a tractable way to study the nonlinear bias.

The procedure is as follows. As in Cinelli and Hazlett (2020), the building blocks are residuals obtained after removing the contribution of the covariates from the outcome

and the treatment. We denote by  $\tilde{D}_i^*$  the standardised residual from regressing  $D$  on  $\mathbf{X}$  (which is the nonlinear analogue of  $\dot{D}_i$ ) and by  $\tilde{Y}_i^*$  the standardised residual from the short nonlinear model (the nonlinear analogue of  $\dot{Y}_i$ ; its precise definition is given below). Unlike in the linear framework, where these residuals are absorbed into the closed-form bias formula and never need to be computed, our approach requires them explicitly.

Once the residuals are in hand, the construction exploits the fact that covariance is a linear operator. We define the synthetic confounder as a linear combination of the available ingredients,

$$Z_i = a \tilde{D}_i^* + b \tilde{Y}_i^* + c \varepsilon_i, \quad \varepsilon_i \sim N(0, 1) \text{ independent of } (\tilde{D}^*, \tilde{Y}^*), \quad (3.3)$$

and choose the loadings  $(a, b, c)$  to deliver the specified correlation coefficients  $\rho_D = \text{Cor}(Z, \tilde{D}^*)$  and  $\rho_Y = \text{Cor}(Z, \tilde{Y}^*)$ . These play the same role as  $R_{D \sim Z|\mathbf{X}}^2$  and  $R_{Y \sim Z|\mathbf{X}}^2$  in the linear framework, measuring how strongly the confounder correlates with the residual variation in treatment and outcome (though our parameters are signed correlations bounded in  $[-1, 1]$ , so that the direction of confounding is part of the specification).

Because  $Z$  is linear in  $(\tilde{D}^*, \tilde{Y}^*, \varepsilon)$  and covariance is linear operator, the two correlation targets translate into a  $2 \times 2$  linear system in  $(a, b)$  that admits a unique closed-form solution. The third coefficient  $c$  is then determined by the normalisation  $\text{Var}(Z) = 1$ , and the requirement  $c^2 \geq 0$  defines the set of correlation pairs  $(\rho_D, \rho_Y)$  that are jointly feasible. The independent noise term  $c \varepsilon_i$  ensures that  $Z$  is not a deterministic function of the residuals, which would cause collinearity in the augmented model. Proposition 3.1 gives the explicit solution and the feasibility condition.

**Proposition 3.1.** *The variable*

$$Z = a \tilde{D}^* + b \tilde{Y}^* + c \varepsilon, \text{ where } \varepsilon \sim i.i.d.N(0, 1) \quad (3.4)$$

*satisfies  $\text{Cor}(Z, \tilde{D}^*) = \rho_D$  and  $\text{Cor}(Z, \tilde{Y}^*) = \rho_Y$  when*

$$a = \frac{\rho_D - \rho_Y r}{1 - r^2}, \quad b = \frac{\rho_Y - \rho_D r}{1 - r^2}, \quad c = \sqrt{1 - a^2 - b^2 - 2abr}. \quad (3.5)$$

The construction is feasible if and only if  $c^2 \geq 0$ , which requires

$$\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Yr \leq 1 - r^2. \quad (3.6)$$

The proof is in Appendix A.5. The feasibility condition (3.6) determines the set of achievable  $(\rho_D, \rho_Y)$  pairs, and it reduces to the unit circle when  $r = 0$ . The quantity  $r$  is a nuisance quantity that guarantees that  $\text{Cor}(Z, \tilde{D}^*) = \rho_D$  and  $\text{Cor}(Z, \tilde{Y}^*) = \rho_Y$  exactly<sup>5</sup>.

Given Proposition 3.1, the synthetic confounder sensitivity analysis is as follows. To fix ideas, let us consider a likelihood model operationalised through a linear index structure (linear index structures is not required for the application of the method), so that

$$\begin{aligned} \mathbb{E}[Y_i \mid D_i, \mathbf{X}_i, Z_i] &= h(\boldsymbol{\alpha}'A_i + \beta D_i + \delta Z_i) \\ &= h(\alpha_0 + \boldsymbol{\gamma}'\mathbf{X}_i + \beta D_i + \delta Z_i), \end{aligned} \quad (3.7)$$

where  $h : \mathbb{R} \rightarrow \mathcal{Y}$  is a known, invertible mean function and  $\boldsymbol{\alpha} = (\alpha_0, \boldsymbol{\gamma})'$  is the nuisance vector from Section 2. Note that this expression does not include an interaction between  $Z$  and  $D$  for convenience, and the extension to relax this homogeneity condition is straightforward, as is discussed below. Writing  $g = h^{-1}$  for the link function, equation (3.7) is equivalent to

$$\eta_i = g(\mu_i) = \alpha_0 + \boldsymbol{\gamma}'\mathbf{X}_i + \beta D_i + \delta Z_i, \quad \mu_i = \mathbb{E}[Y_i \mid D_i, \mathbf{X}_i, Z_i]. \quad (3.8)$$

---

<sup>5</sup>The  $r$  is 0 or near zero for a large family of models. For example, for GLM, letting  $e_i = Y_i - \hat{\mu}_i$  be the raw prediction error from the short model, and  $\tilde{D}_i$  be the OLS treatment residual. The MLE satisfies the normal equations  $\sum_i (Y_i - \hat{\mu}_i)D_i = 0$  and  $\sum_i (Y_i - \hat{\mu}_i)X_{ij} = 0$  for all  $j$ . Since  $\tilde{D}_i = D_i - \sum_j X_{ij}\hat{\beta}_j$  is a linear combination of  $D_i$  and  $\{X_{ij}\}_{j=1}^p$ , then

$$\sum_i e_i \tilde{D}_i = \sum_i (Y_i - \hat{\mu}_i)D_i - \sum_j \hat{\beta}_j \sum_i (Y_i - \hat{\mu}_i)X_{ij} = 0$$

In practice we rely on standardised residuals. Dividing by  $\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}$  introduces observation-specific weights that break exact orthogonality, so  $r = \text{Cor}(\tilde{D}^*, \tilde{Y}^*)$  is approximately but not exactly zero (of order  $10^{-4}$  in well-specified models). For this reason, in practice we compute  $r$  from the data rather than fixing it to zero. This is particularly to improve the precision of the bisection method described in later sections.

This single-index structure encompasses all GLMs (logit, probit, Poisson, negative binomial), Tobit, ordered probit, and the fractional response model of Papke and Wooldridge (1996), among others. The parameter of interest is the average marginal effect:

$$\overline{\text{AME}} = \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial D_i} \mathbb{E}[Y_i | D_i, \mathbf{X}_i, Z_i] = \frac{1}{n} \sum_{i=1}^n \beta h'(\eta_i). \quad (3.9)$$

Because  $Z_i$  is unobservable, the researcher estimates the short model

$$\eta_i^* = g(\mu_i^*) = \alpha_0^* + \boldsymbol{\gamma}^{*\prime} \mathbf{X}_i + \beta^* D_i. \quad (3.10)$$

Let  $\hat{\beta}^*$  denote the short-model MLE, and let the short-model AME be  $\widehat{\overline{\text{AME}}}^* = n^{-1} \sum_i \hat{\beta}^* h'(\hat{\eta}_i^*)$ . The objective of the sensitivity analysis is to characterise how  $\widehat{\overline{\text{AME}}}^*$  changes under the influence of confounders of different strengths.

For a user-specified grid of  $(\rho_D, \rho_Y)$  pairs:

1. Compute  $a$ ,  $b$ ,  $c$  from (3.5); check feasibility via (3.6).
2. Draw  $\varepsilon \sim N(0, 1)$ .
3. Construct  $Z$  via (3.4) and augment the short model with  $Z$  as an additional regressor.
4. Re-estimate the long model (3.8) by maximum likelihood and compute the adjusted AME (3.9).

The above procedure allows us map each feasible  $(\rho_D, \rho_Y)$  to an adjusted AME. If the mapping excludes zero for a given  $\rho_D$ , the estimated effect is judged robust to confounding of that strength. Because the full model is re-estimated numerically with the constructed  $Z$ , no Taylor approximation is required and the bias attributable to this specific confounder is computed exactly. The sensitivity surface therefore reflects the bias from one representative confounder at each  $(\rho_D, \rho_Y)$  pair, rather than a class-wide bound.

### 3.1 Construction of residuals.

The discussion so far assumed the existence of suitable standardized residuals. In practice, these need to be computed as a first step in such a way that they capture variation unexplained by  $\mathbf{X}$  (and  $D$ ) in the short model. The natural choice in each model family is the standardised residual

$$\tilde{Y}_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}, \quad (3.11)$$

where  $\hat{\mu}_i$  is the short-model fitted value and  $V(\cdot)$  is the variance function of the model family. We give two examples next (a third example is discussed in the following section).

**Example 3.1.** Consider first a binary logit, which gives fitted probabilities  $\hat{\mu}_i = \Lambda(\hat{\eta}_i^*)$  where  $\Lambda$  is the logistic CDF and  $\hat{\eta}_i^* = \hat{\alpha}_0^* + \hat{\beta}^* D_i + \hat{\gamma}^{*\prime} \mathbf{X}_i$ . The variance function is  $V(\mu) = \mu(1 - \mu)$ , so the standardised residual is

$$\tilde{Y}_i = \frac{Y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}}. \quad (3.12)$$

which is already scaled by the conditional standard deviation of  $Y_i$ . This residual is then used in Proposition 3.1.

**Example 3.2.** Consider, next, the standard Tobit model with latent outcome  $Y_i^* = \alpha_0 + \beta D_i + \gamma' \mathbf{X}_i + u_i$ ,  $u_i \sim N(0, \sigma^2)$ , and observed outcome  $Y_i = \max(0, Y_i^*)$ . The conditional mean and variance of the observed censored outcome are

$$\hat{E}[Y_i | \mathbf{X}_i] = \Phi(\hat{z}_i)(\hat{\alpha}_0 + \hat{\beta} D_i + \hat{\gamma}' \mathbf{X}_i + \hat{\sigma} \hat{\lambda}_i), \quad (3.13)$$

$$\widehat{Var}[Y_i | \mathbf{X}_i] = \hat{\sigma}^2 \Phi(\hat{z}_i)[1 - \Phi(\hat{z}_i)(\hat{z}_i + \hat{\lambda}_i)^2], \quad (3.14)$$

where  $\hat{z}_i = (\hat{\alpha}_0 + \hat{\beta} D_i + \hat{\gamma}' \mathbf{X}_i)/\hat{\sigma}$  and  $\hat{\lambda}_i = \phi(\hat{z}_i)/\Phi(\hat{z}_i)$  is the inverse Mills ratio. The standardised residual for Tobit is therefore

$$\tilde{Y}_i = \frac{Y_i - \hat{E}[Y_i | \mathbf{X}_i]}{\sqrt{\widehat{Var}[Y_i | \mathbf{X}_i]}}. \quad (3.15)$$

### 3.2 Treatment Residuals

Our method also requires the treatment residuals. We can use the standard OLS residual  $\hat{D}_i = D_i - \mathbf{X}'_i \hat{b}^{\text{OLS}}$ ,  $\hat{b}^{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'D$ , which is then standardised to zero mean and unit variance to obtain

$$\tilde{D}_i^* = \frac{\hat{D}_i - \bar{\hat{D}}}{\hat{\sigma}_{\hat{D}}},$$

where  $\bar{\hat{D}}$  and  $\hat{\sigma}_{\hat{D}}$  are the sample mean and standard deviation of  $\hat{D}_i$ . It is  $\tilde{D}_i^*$  that enters Proposition 3.1 as the treatment residual primitive. When  $D$  is binary one might consider replacing the OLS residual  $\hat{D}_i$  with a probit or logit residual  $D_i - \Phi(\mathbf{X}'_i \hat{\gamma})$  before standardising. However, the construction of  $Z$  in Proposition 3.1 is valid for any choice of treatment residual prior to standardisation. Specifically, the loadings  $a, b, c$  adjust automatically through (3.5), so that the target correlations  $\rho_D$  and  $\rho_Y$  are achieved exactly regardless of which residual is standardised. The choice only affects the interpretation of  $\rho_D$ . In practice we use the OLS residual throughout for two reasons. First,  $\rho_D$  then matches the Cinelli–Hazlett partial correlation exactly via the linear FWL identity, facilitating benchmarking against that framework. Second, for moderate propensity scores the OLS and probit residuals are nearly proportional, so the choice has little practical impact.

### 3.3 Interpretation of partial correlations.

The partial correlations  $(\rho_D, \rho_Y)$  have a direct interpretation in terms of the amount of residual variation that the synthetic confounder is allowed to explain. By construction,  $\tilde{D}^*$  and  $\tilde{Y}^*$  are scalar, standardised residuals. Thus, once the short model has been fitted,  $\rho_Y = \text{Cor}(Z, \tilde{Y}^*)$  is simply the correlation between the confounder and the part of the outcome that remains unexplained by  $D$  and  $\mathbf{X}$ . Similarly,  $\rho_D = \text{Cor}(Z, \tilde{D}^*)$  measures the association between the confounder and the part of the treatment that remains unexplained by  $\mathbf{X}$ .

Because these are ordinary correlations between two scalar variables, the usual relationship  $R^2 = \rho^2$  applies. In particular, if one were to regress  $\tilde{Y}^*$  on  $Z$ , the resulting  $R^2$

would be exactly  $\rho_Y^2$ . Hence  $\rho_Y$  can be interpreted as specifying how much of the residual variation in the outcome the confounder is allowed to explain. For example, choosing  $\rho_Y = 0.3$  means that the confounder explains 9% of the variance in the outcome residual. Likewise, choosing  $\rho_D = 0.2$  means that the confounder explains 4% of the residual variation in the treatment after  $\mathbf{X}$  has been partialled out.

### 3.4 Homogeneity

The single-index structure (3.7) assumes that the confounder  $Z_i$  enters the outcome equation additively with a homogeneous coefficient  $\delta$ , ruling out interactions between  $Z_i$  and the treatment  $D_i$ . This assumption is made for expositional convenience, but it is not built into the way the confounder itself is constructed.  $Z$  uses only the residual variation in the treatment and outcome from the short model. The coefficients  $(a, b, c)$  are chosen to make  $Z$  have the researcher-specified correlations  $(\rho_D, \rho_Y)$  with  $\tilde{D}$  and  $\tilde{Y}$ . Nothing in this construction specifies how  $Z$  subsequently affects the outcome. Said differently, the parameters  $(\rho_D, \rho_Y)$  define how strongly is the confounder related to the treatment and the unexplained part of the outcome, but say nothing about the confounder affects the outcome.

If treatment-effect heterogeneity is of interest, the researcher can therefore augment the re-estimated model with an interaction between the synthetic confounder and the treatment:

$$g(\mu_i) = \alpha_0 + \boldsymbol{\gamma}'\mathbf{X}_i + \beta D_i + \delta Z_i + \phi \{Z_i \times D_i\}, \quad (3.16)$$

and re-estimate (3.16) by maximum likelihood in place of (3.16). The adjusted AME from (3.16) then incorporates a confounder whose effect on the outcome is allowed to vary with the level of  $D_i$ .

## 4 Fragility Thresholds and Benchmarking

The method described in the previous section allows researchers to fix the specific values of the partial correlations, enabling the characterisation of a sensitivity surface. In practice,

however, one wants to know how strong the confounding must be to overturn a result. In this regard, Cinelli and Hazlett (2020), Masten and Poirier (2022) and Diegert et al. (2025), note the existence of two natural thresholds to consider, namely the explain-away threshold and the sign change threshold.

Consider a fixed level of  $\rho_Y$  and define  $g(\rho_D) = \widehat{\text{AME}}(\rho_D, \rho_Y)$ . The feasibility condition (3.6) defines a feasible range of variation for  $\rho_D$ . Setting  $c^2 = 0$  and solving for  $\rho_D$ , we obtain two roots,

$$\rho_D^{\pm \text{bdy}} = r\rho_Y \pm \sqrt{(1-r^2)(1-\rho_Y^2)}. \quad (4.1)$$

Within the feasible interval defined by these roots,  $[\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}]$ , we can first define the explain-away threshold

$$\rho_D^*(\rho_Y) = \arg \min_{|\rho_D|} \left\{ \rho_D \in [\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}] : |g(\rho_D) / \widehat{\text{SE}}(\rho_D)| < z_{\alpha/2} \right\} \quad (4.2)$$

where  $\widehat{\text{SE}}(\rho_D)$  is the standard error of the AME. In other words,  $\rho_D^*(\rho_Y)$  constitutes the minimal amount of confounding required to overturn the statistical significance of AME. Similarly, the sign-change threshold

$$\rho_D^+(\rho_Y) = \arg \min_{|\rho_D|} \left\{ \rho_D \in [\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}] : \text{sign}(g(\rho_D)) \neq \text{sign}(g(0)) \right\} \quad (4.3)$$

is the feasible value closest to zero at which the direction of the estimated effect reverses.

Computing  $\rho_D^*$  and  $\rho_D^+$  requires searching over the feasible interval. This can be done with standard numerical methods. In practice, we rely on a grid-search followed by a bisection to compute the values. More specifically, in a first stage, the synthetic method is applied on a grid of equally-spaced points within  $[\rho_D^{-\text{bdy}}, \rho_D^{+\text{bdy}}]$ . This search identifies an initial pair of grid points where the criterion switches from unmet to met. Within that interval we can apply a standard bisection algorithm to find the root of the function. If no grid point triggers the criterion, the result is reported as robust for that threshold<sup>6</sup>.

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<sup>6</sup>It is convenient to emphasise here that in the implementation of the algorithm  $\varepsilon$  in (3.4) is fixed across all grid evaluations. This ensures that  $g(\rho_D)$  varies smoothly as a function of  $\rho_D$ , which makes both the grid search reliable and the bisection refinement well-behaved.

## 4.1 Benchmarking

As described so far, the synthetic confounder approach (as the Cinelli and Hazlett, 2020 framework) is subject to two main limitations. First, the values of  $\rho_Y$  (and  $\rho_D$ ) are pre-specified by the researcher and thus subject to some subjectivity (and potentially detached from the specific application at hand). Second, and more importantly, the fragility thresholds  $\rho_D^*$  and  $\rho_D^+$  measure the confounding strength required to render a result statistically insignificant or to reverse its sign, yet a heavily biased result can appear robust precisely because the bias inflates the initial  $t$ -statistic<sup>7</sup>.

The solution proposed by Cinelli and Hazlett (2020) for both problems is benchmarking. Rather than asking whether any hypothetical confounder can explain away the result, benchmarking asks how much the estimate would shift if a confounder as strong as a specific observed variable had been omitted. By using an observable regressor as a benchmark, we are grounding the choice of  $(\rho_D, \rho_Y)$  in the data and resolving the subjectivity problem. This also addresses the inflated-baseline problem. If the researcher can identify an observed covariate whose partial correlations with treatment and outcome are at least as large as those of the true omitted confounder, then the benchmark-implied correction will be large enough to reveal if the short-model estimate is substantially driven by confounding, regardless of how inflated the initial  $t$ -statistic is. Note, however, that the credibility of any ensuing conclusion depends on the choice of benchmark.

As in the Cinelli–Hazlett framework, the benchmarking procedure for likelihood based models computes  $(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$  for each candidate variable  $W_k$ , and then it evaluates the adjusted AME at those levels. Comparing  $\hat{\rho}_{D,k}$  to  $\rho_D^*(\hat{\rho}_{Y,k})$  one can explore whether a confounder of  $W_k$ ’s strength would explain away the result, while the adjusted AME and implied bias quantify how much the short-model estimate would be overstated.

More precisely, the procedure is implemented as follows. Let  $\mathbf{X}_{-k}$  denote the control vector with  $W_k$  removed. Then:

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<sup>7</sup>To see this last point, note that the explain-away threshold  $\rho_D^*$  tells us how large must  $\rho_D$  be to drive  $|t(\rho_D)| < z_{\alpha/2}$ . Since  $t(\rho_D) = \widehat{\text{AME}}(\rho_D) / \widehat{\text{SE}}(\rho_D)$ , a large threshold means that the adjusted AME must fall a long way before it becomes insignificant. But if the short model AME is itself inflated by omitted variable bias, the starting point is artificially high, and the distance to insignificance is correspondingly large. The threshold is therefore measuring distance from a biased baseline, not distance from the truth.

1. Regress  $W_k$  on  $\mathbf{X}_{-k}$  by OLS and save the standardised residuals

$$W_{r,k} = \frac{W_k - \mathbf{X}_{-k}\hat{b}}{\text{SD}(W_k - \mathbf{X}_{-k}\hat{b})}.$$

This partialling-out removes variation in  $W_k$  that is collinear with the remaining controls, so that the subsequent correlations measure the partial association of  $W_k$  with treatment and outcome after conditioning on everything else in  $\mathbf{X}$ .

2. Re-estimate the OLS regression of  $D$  on  $\mathbf{X}_{-k}$  (excluding  $W_k$ ) and save the standardised residuals  $\tilde{D}_{-k}$ . Compute

$$\hat{\rho}_{D,k} = \text{Cor}(W_{r,k}, \tilde{D}_{-k}).$$

3. Re-estimate the same nonlinear model used for the main analysis,  $\text{model}(Y) \sim D + \mathbf{X}_{-k}$ , and compute standardised residuals  $\tilde{Y}_{-k}$  from the fitted values of this reduced model. For binary, ordered, and count models these are

$$\tilde{Y}_{-k,i} = \frac{Y_i - \hat{\mu}_{-k,i}}{V(\hat{\mu}_{-k,i})^{1/2}},$$

where  $V(\cdot)$  is the variance function of the model family (e.g.  $\hat{\mu}(1 - \hat{\mu})$  for logit and probit,  $\hat{\mu}$  for Poisson,  $\hat{\mu} + \hat{\phi}\hat{\mu}^2$  for negative binomial). Compute

$$\hat{\rho}_{Y,k} = \text{Cor}(W_{r,k}, \tilde{Y}_{-k}),$$

or  $\hat{\rho}_{Y,k,j} = \text{Cor}(W_{r,k}, \tilde{Y}_{-k,j})$  for each outcome  $j$  in the mlogit case.

4. Check feasibility of  $(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$  via the loading formulas (3.5). If feasible, construct  $Z$  using the main-model residuals  $\tilde{D}$  and  $\tilde{Y}$ , fit the long model, and record the adjusted AME, SE,  $t$ -statistic, and

$$\text{bias} = \widehat{\text{AME}}_{\text{short}} - \widehat{\text{AME}}(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k}).$$

A large bias indicates that omitting a confounder as influential as  $W_k$  would materially distort the estimated treatment effect.

The benchmark output reports, for each variable  $W_k$ ,  $\hat{\rho}_{D,k}$  and  $\hat{\rho}_{Y,k}$  can be compared to the bisection thresholds. If  $\hat{\rho}_{D,k} < \rho_D^*(\hat{\rho}_{Y,k})$ , the benchmarking variable's confounding strength falls short of the explain-away threshold. We would then conclude that even if such a variable were omitted, the result would remain statistically significant. Conversely, if  $\hat{\rho}_{D,k} > \rho_D^*(\hat{\rho}_{Y,k})$ , the omission of  $W_k$  would have been enough to explain away the result.

## 5 Simulation Evidence

This section presents a simulation study aimed at evaluating the performance of the method. The study consists of 3 parts. First we study behaviour under different confounding. Next we verify the correct functioning of the benchmarking procedure and explore its limitations. Finally, we explore the advantages of the synthetic method over the exact method in Cinelli and Hazlett (2020) when the underlying DGP is not linear.

### 5.1 Sensitivity analysis under varying confounding

The first study evaluates the performance of the synthetic confounder under different confounding scenarios. We drew samples of  $n = 20,000$  observations from four different data generating process (DGPs). In each DGP, the outcome follows a binary logit model

$$Y \sim \text{logit}(-0.5 + \beta_0 D + 0.3X_1 + 0.2X_2 + \delta_0 Z) \quad (5.1)$$

where  $X_j \sim N(0, 1)$ ,  $j = 1, 2$ . The treatment is  $D = \mathbf{1}(0.3X_1 + u > 0)$ ,  $u \sim N(0, 1)$ . The confounder,  $Z$  is constructed as  $Z = \rho_D^{\text{true}} \tilde{D} + \sqrt{1 - (\rho_D^{\text{true}})^2} \eta$ , where  $\eta \sim N(0, 1)$ , so that  $\text{Cor}(Z, \tilde{D}) = \rho_D^{\text{true}}$  by construction. The variables  $X_j, u$  and  $\eta$  are i.i.d. We considered the following scenarios:

- DGP 1, No confounder:  $\beta_0 = 0.5, \delta_0 = 0$
- DGP 2, Weak confounder:  $\beta_0 = 0.5, \delta_0 = 0.2, \rho_D^{\text{true}} = 0.3 \rightarrow \rho_Y^{\text{true}} \approx 0.09$ .

- DGP 3, Strong confounder:  $\beta_0 = 0.3$ ,  $\delta_0 = 0.8$ ,  $\rho_D^{\text{true}} = 0.6 \rightarrow \rho_Y^{\text{true}} \approx 0.22$ .
- DGP 4, Spurious significance:  $\beta_0 = 0$ ,  $\delta_0 = 0.8$ ,  $\rho_D^{\text{true}} = 0.4 \rightarrow \rho_Y^{\text{true}} \approx 0.29$

**Table 1:** True AMEs, short-model estimates, and confounding bias by DGP

| DGP                       | $\beta_D$ | $\delta_Z$ | $\rho_D^{\text{true}}$ | True AME | Short-model AME | Bias    |
|---------------------------|-----------|------------|------------------------|----------|-----------------|---------|
| 1 (No confounder)         | 0.5       | 0.0        | 0.5                    | 0.1170   | 0.1101          | −0.0068 |
| 2 (Weak confounder)       | 0.5       | 0.2        | 0.3                    | 0.1153   | 0.1335          | 0.0183  |
| 3 (Strong confounder)     | 0.3       | 0.8        | 0.6                    | 0.0608   | 0.2650          | 0.2041  |
| 4 (Spurious significance) | 0.0       | 0.8        | 0.4                    | 0.0000   | 0.1227          | 0.1227  |

Notes:  $\beta_D$  is the logit coefficient on  $D$  in the true model,  $\delta_Z$  is the coefficient on the omitted confounder  $Z$ . The true AME is estimated using a  $n = 500,000$  simulated sample. The bias = Short-model AME − True AME.

Table 1 describes the properties of short-model estimates under these scenarios<sup>8</sup> alongside the short-model estimates. There is no confounding in DGP 1 and, as a result, the short model AME in Table 1 is only subject to sampling noise. In DGP 2, the confounder has a small direct effect on the outcome and moderate association with treatment, producing a modest upward bias of 0.0183 (14%). In DGP 3, the treatment coefficient is small relative to the confounder creating a scenario of severe upward bias. The short-model AME of 0.265 overstates the true AME four-fold. DGP 4 is the most damaging scenario from a policy and research perspective. The true treatment effect is zero, but the strong confounder creates a significant ( $t \approx 18$ ) short-model AME of 0.123.

Table 2 reports the fragility thresholds for all four DGPs. Each element in Table 2 measures, for an omitted confounder that has outcome-residual association  $\rho_Y$ , how strongly would the confounder need to be correlated with treatment before the result is explained away or changes sign. A small threshold indicates that even a weak confounder would suffice. In the table, “robust” indicates that no feasible confounder of that type can overturn the result.

Reading across DGPs at any fixed  $\rho_Y > 0$ , the thresholds for DGPs 1, 2 and 4 are relatively similar, while the results obtained with DGP 3 requires a strong confounder to be wiped out. As discussed in Section 4.1, the latter reflects the inflated  $t$ -statistic

<sup>8</sup>For each DGP, the true AME and the implied  $\rho_Y^{\text{true}}$  are computed from a simulated sample of  $n = 500,000$  draws from the true DGP

**Table 2:** Fragility thresholds by DGP and  $\rho_Y$ .

| $\rho_Y$ | DGP 1      |            | DGP 2      |            | DGP 3      |            | DGP 4      |            |
|----------|------------|------------|------------|------------|------------|------------|------------|------------|
|          | $\rho_D^*$ | $\rho_D^+$ | $\rho_D^*$ | $\rho_D^+$ | $\rho_D^*$ | $\rho_D^+$ | $\rho_D^*$ | $\rho_D^+$ |
| −0.5     | −0.312     | −0.361     | −0.411     | −0.471     | robust     | robust     | −0.363     | −0.415     |
| −0.4     | −0.352     | −0.414     | −0.478     | −0.564     | robust     | robust     | −0.410     | −0.481     |
| −0.3     | −0.450     | −0.557     | −0.677     | robust     | robust     | robust     | −0.534     | −0.675     |
| −0.2     | robust     | robust     | robust     | robust     | robust     | robust     | robust     | robust     |
| −0.1     | robust     | robust     | robust     | robust     | robust     | robust     | robust     | robust     |
| 0.0      | 0.980      | 0.997      | 0.983      | 0.996      | 0.993      | robust     | 0.995      | robust     |
| 0.1      | 0.583      | 0.630      | 0.641      | 0.678      | 0.809      | 0.823      | 0.637      | 0.677      |
| 0.2      | 0.396      | 0.437      | 0.456      | 0.492      | 0.678      | 0.694      | 0.443      | 0.481      |
| 0.3      | 0.295      | 0.327      | 0.348      | 0.377      | 0.577      | 0.593      | 0.331      | 0.363      |
| 0.4      | 0.234      | 0.260      | 0.279      | 0.303      | 0.500      | 0.514      | 0.263      | 0.288      |
| 0.5      | 0.196      | 0.216      | 0.232      | 0.252      | 0.439      | 0.452      | 0.218      | 0.238      |

$\rho_D^*$  is the explain away threshold (the point at which the adjusted AME becomes statistically insignificant;  $|t| < 1.96$ ).  $\rho_D^+$  is the threshold at which the sign of the adjusted AME reverses. “robust” indicates that no threshold is reached within the feasible range.

in DGP 3 rather than robustness. The confounding bias has made the short-model result look so significant that even a powerful synthetic confounder cannot drive it to insignificance without very high  $\rho_D$ . Within any column, the thresholds fall monotonically as  $\rho_Y$  increases from 0 to 0.5, because a confounder more strongly associated with the outcome needs a smaller treatment association to cause bias. At  $\rho_Y = 0$  the threshold is near the feasibility boundary for all DGPs, because a confounder that is irrelevant for the outcome cannot bias the treatment effect estimate regardless of its association with treatment.

The comparison between DGPs 1 and 4 is the most relevant for practitioners. Their thresholds are similar at every  $\rho_Y$ . However, while DGP 1 has a true positive treatment effect DGP 4 has a true AME of zero. The threshold table cannot distinguish these two cases. A spurious result with a moderate  $t$ -statistic looks identical to a true, significant result. This is the main limitation of significance-based fragility thresholds and motivates the benchmark analysis in Cinelli and Hazlett (2020) and in Section 4.1.

Note further that in DGPs 1, 2, and 3, the true confounder has a positive effect on the outcome ( $\delta_Z > 0$ ) and a positive association with treatment ( $\rho_D^{\text{true}} > 0$ ), but the true treatment effect is also positive. At the row of the table closest to the true  $\rho_Y$  of each DGP, the reported explain-away threshold  $\rho_D^*$  exceeds the true  $\rho_D^{\text{true}}$ . This is expected

because although the true confounder exists and creates bias, it is not strong enough to fully explain away the result (since there is a non-trivial treatment effect left after adjusting for it). The method correctly identifies that confounding at the true level is insufficient to overturn the conclusion. Conversely, in DGP 4, the true treatment effect is zero and the short-model AME is an artifact. At the row closest to the true  $\rho_Y$ , the reported threshold  $\rho_D^*$  falls below the true  $\rho_D^{\text{true}}$ . This means the true confounder is strong enough to explain away the result. Finally, note that across all four DGPs,  $|\rho_D^+| > |\rho_D^*|$ , as expected.

## 5.2 Study 2: Benchmark validation

We next evaluate the benchmark procedure, using the same DGPs as above. We begin by verifying the correct functioning of the procedure by using the true confounder  $Z$  as the benchmark variable. The procedure should then correctly identify the empirical confounding strength of  $Z$  and the implied adjusted AME should recover the true value. Table 3 reports results for all four DGPs.

**Table 3:** Benchmark analysis by DGP ( $n = 20,000$ ).

| DGP | Var.  | $\hat{\rho}_D$ | $\hat{\rho}_Y$ | ME     | SE    | $t$   | Correction |
|-----|-------|----------------|----------------|--------|-------|-------|------------|
| 1   | $X_1$ | 0.233          | 0.144          | 0.074  | 0.007 | 10.57 | 0.036      |
|     | $X_2$ | -0.005         | 0.081          | 0.111  | 0.007 | 16.35 | -0.001     |
|     | $Z$   | 0.495          | -0.002         | 0.107  | 0.008 | 13.51 | 0.003      |
| 2   | $X_1$ | 0.224          | 0.136          | 0.100  | 0.007 | 14.52 | 0.034      |
|     | $X_2$ | -0.004         | 0.101          | 0.134  | 0.007 | 20.13 | -0.001     |
|     | $Z$   | 0.298          | 0.095          | 0.100  | 0.007 | 14.10 | 0.034      |
| 3   | $X_1$ | 0.237          | 0.077          | 0.243  | 0.006 | 40.58 | 0.022      |
|     | $X_2$ | 0.015          | 0.082          | 0.264  | 0.006 | 46.37 | 0.001      |
|     | $Z$   | 0.603          | 0.222          | 0.057  | 0.008 | 7.43  | 0.208      |
| 4   | $X_1$ | 0.232          | 0.089          | 0.102  | 0.007 | 14.57 | 0.021      |
|     | $X_2$ | -0.015         | 0.087          | 0.124  | 0.007 | 18.44 | -0.002     |
|     | $Z$   | 0.401          | 0.293          | -0.014 | 0.007 | -1.98 | 0.137      |

ME is the adjusted AME evaluated at  $(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$ . "correction" is defined as  $\widehat{\text{AME}}_{\text{short}} - \widehat{\text{AME}}(\hat{\rho}_{D,k}, \hat{\rho}_{Y,k})$ . The true AMEs are 0.117 (DGP 1), 0.115 (DGP 2), 0.06 (DGP 3) and 0 (DGP 4). Short-model AMEs are: 0.110 (DGP 1), 0.134 (DGP 2), 0.265 (DGP 3), 0.123 (DGP 4).  $Z$  denotes the true omitted confounder included only for validation.

In DGP 1, (where  $Z$  has no effect on  $Y$ ) the  $Z$  benchmark yields  $\hat{\rho}_{Y,Z} = -0.002$  and

an implied bias of 0.003. The adjusted AME of 0.107 is essentially equal to the short-model estimate of 0.110, correctly indicating negligible confounding despite the strong treatment association ( $\hat{\rho}_{D,Z} = 0.495$ ). In DGP 2, the  $Z$  benchmark gives  $\hat{\rho}_{D,Z} = 0.298$  and  $\hat{\rho}_{Y,Z} = 0.095$ , consistent with the DGP design ( $\rho_D^{\text{true}} = 0.3$ , small outcome effect  $\delta_Z = 0.2$ ). The implied bias (correction) of 0.034 accounts for roughly one quarter of the short-model AME of 0.134. The adjusted  $t$ -statistic of 14.1 confirms the result survives confounding at  $Z$ 's empirical strength. Comparing  $\hat{\rho}_{D,Z} = 0.298$  to the explain-away threshold at  $\rho_Y \approx 0.1$  from Table 2 ( $\rho_D^* = 0.641$ ), the benchmark point lies inside the robust region.

DGP 3 and DGP 4 are perhaps the most important ones in terms of validating the synthetic confounder method. In DGP 3, the  $Z$  benchmark gives  $\hat{\rho}_{D,Z} = 0.603$  and  $\hat{\rho}_{Y,Z} = 0.222$ , both closely matching the DGP design values ( $\rho_D^{\text{true}} = 0.6$ ,  $\delta_Z = 0.8$ ). The correction is 0.208, reducing the adjusted AME from 0.265 to 0.057, which is close to the true AME of 0.061 from Table 1. The adjusted  $t$ -statistic of 7.43 confirms the result is still statistically significant at the true confounding strength. This is consistent with the explain-away threshold at  $\rho_Y = 0.2$  from Table 2 ( $\rho_D^* = 0.678$ ), and since  $\hat{\rho}_{D,Z} = 0.603 < 0.678$ , the benchmark correctly predicts the result is not explained away even by the true confounder.

In DGP 4, the  $Z$  benchmark gives  $\hat{\rho}_{D,Z} = 0.401$  and  $\hat{\rho}_{Y,Z} = 0.293$ , closely matching the DGP design ( $\rho_D^{\text{true}} \approx 0.4$ ,  $\delta_Z = 0.8$ ). The correction is 0.137, reducing the adjusted AME from 0.123 to  $-0.014$  with  $t = -1.98$  (essentially losing statistical significance at 5% -up to some statistical noise). The adjusted AME thus indicates that the significant short-model result is spurious. Comparing  $\hat{\rho}_{D,Z} = 0.401$  to the explain-away threshold at  $\rho_Y \approx 0.3$  from Table 2 ( $\rho_D^* = 0.331$ ), the benchmark values exceed the threshold, correctly predicting that a confounder as strong as  $Z$  would explain away the result which the adjusted  $t = -1.98$  directly confirms.

In practice, the researcher does not observe the true confounder and must rely on  $X_1$  and  $X_2$  as benchmarks. In our DGPs,  $X_2$  is orthogonal to treatment by construction and is therefore a poor benchmark. Unsurprisingly, then, after partialling out  $X_1$ ,  $\hat{\rho}_{D,X_2}$  is

close to zero in all four DGPs (  $-0.005$ ,  $-0.004$ ,  $0.015$ ,  $-0.015$  respectively). The covariate  $X_2$  does not yield any corrections, regardless of  $\hat{\rho}_{Y,X_2}$ . Even though  $X_2$  has a moderate association with the outcome residual ( $\hat{\rho}_{Y,X_2} \approx 0.08$ – $0.10$  across DGPs), omitting it would not cause confounding bias because it is not related to treatment.

The covariate  $X_1$  is moderately correlated with treatment in all four DGPs ( $\hat{\rho}_{D,X_1} \approx 0.22$ – $0.24$ ) and also with the outcome residual ( $\hat{\rho}_{Y,X_1} \approx 0.08$ – $0.14$ ). In low confounding scenarios (DGPs 1 and 2)  $X_1$  benchmark correctly indicates that confounding at its level would not overturn the result. The adjusted  $t$ -statistics remain significant (  $10.6$  and  $14.5$  respectively). In DGP 2, the  $X_1$  benchmark gives almost identical results to the  $Z$  benchmark (bias  $0.034$  in both cases). In DGP 3, the  $X_1$  benchmark leaves the result highly significant ( $t = 40.6$ , bias =  $0.022$ ), while the true confounder  $Z$  has  $\hat{\rho}_{D,Z} = 0.603$  and  $\hat{\rho}_{Y,Z} = 0.222$ . Note, however, that  $X_1$ 's benchmark is weak so it could only generate a small correction regardless of the true confounding. Therefore, the conclusion that the result survives is correct, however it is weakly justified.

Once again, DGP 4 is the most important case, as it illustrates the limitations of observed-covariate benchmarking. The  $X_1$  and  $X_2$  benchmarks yield adjusted  $t$ -statistics of  $14.6$  and  $18.4$  respectively thus suggesting the result is robust to confounding at the level of any observed control. A researcher with access only to  $X_1$  and  $X_2$  as benchmarks would have no reason to doubt the result. Thus, overall we see that if the true confounder has a stronger association with the outcome than any observed control, then the benchmark may fail to detect confounding.

The results obtained in DGP 3 and DGP 4 simply emphasise the limitations of sensitivity analysis in general. A benchmark can correctly signal robustness while still understating the extent of confounding-driven inflation (DGP 3), or fail to flag a spurious result entirely if no observed covariate approaches the true confounder's strength (DGP 4). From this we conclude, first, that subject-matter knowledge is critical to assess whether plausible omitted variables could be stronger confounders than any covariate in the model and, second, that benchmarking provides reliable bias correction only when the true confounder is no stronger than the best available benchmark on both the treatment

and outcome margins simultaneously. When the true confounder exceeds the observed benchmarks, the benchmark-implied correction will understate the bias, and the result may appear more robust than it truly is.

### 5.3 Study 3: Comparison with Cinelli–Hazlett

Section 2 showed that the nonlinear OVB may be approximated by the Cinelli and Hazlett (2020) framework in low confounding settings, and we discussed an adapted version of that approach inspired by the slope-coefficient interpretation of the leading term of the OVB. We next evaluate the relative performance of the synthetic confounder method compared to the direct application of the Cinelli–Hazlett framework (CH-OLS) and to the adapted version (CH-adapted) described in Section 2.

Cinelli and Hazlett (2020) show that, in a linear model, the bias in the short-model OLS estimator due to omitting  $Z$  can be written as a function of  $R_{D \sim Z | \mathbf{X}}^2$  (the share of residual treatment variation explained by  $Z$ ), and  $R_{Y \sim Z | D, \mathbf{X}}^2$  (the share of residual outcome variation explained by  $Z$  after conditioning on treatment). The bias formula is

$$|\text{bias}| = \sqrt{\frac{R_{Y \sim Z | D, \mathbf{X}}^2 \cdot R_{D \sim Z | \mathbf{X}}^2}{1 - R_{D \sim Z | \mathbf{X}}^2}} \cdot \widehat{\text{SE}}(\hat{\beta}^*) \cdot \sqrt{df}, \quad (5.2)$$

and the adjusted estimate is  $\hat{\beta}_{\text{adj}}^* = \hat{\beta}^* - \text{sign}(\hat{\beta}^*) \cdot |\text{bias}|$ . In the linear model,  $R_{D \sim Z | \mathbf{X}}^2 = \rho_{D,Z}^2$  and  $R_{Y \sim Z | D, \mathbf{X}}^2 = \rho_{Y,Z}^2$ , where  $\rho_{D,Z}$  and  $\rho_{Y,Z}$  denote the partial correlations of  $Z$  with the treatment and outcome residuals after partialling out  $\mathbf{X}$ . For nonlinear models, Lemma A.2 shows that an additional remainder  $R = O(\delta_0^2)$  enters the bias expression, so (5.2) becomes a first-order approximation whose error grows quadratically with confounding strength.

Because  $Z$  is not observed in practice, a researcher specifies hypothetical values of the two  $R^2$  quantities and benchmarks them against the partial  $R^2$  of observed covariates to assess whether such values are plausible. In the present simulation  $Z$  is observed by construction, which allows us to compute the true benchmark correlations  $\hat{\rho}_{D,Z}$  and  $\hat{\rho}_{Y,Z}$  in each replication. We can then let  $R_{DZ}^2 = \hat{\rho}_{D,Z}^2$  and  $R_{YZ}^2 = \hat{\rho}_{Y,Z}^2$  and use these

quantities to all three methods. It would follow that any difference in the quality of the adjusted AME across methods is attributable to the methods themselves, rather than benchmarking.

Specifically, the methods are as follows:

1. In the standard Cinelli-Hazlett method (CH-OLS) the binary outcome is approximated by a linear probability model of  $Y$  on  $D$  and  $\mathbf{X}$ , computes OLS residuals for  $Y$  and  $D$ , sets  $R_{DZ}^2 = \hat{\rho}_{D,Z}^2$  and  $R_{YZ}^2 = \hat{\rho}_{Y,Z}^2$  from those residuals, and plugs into (5.2) to obtain an adjusted coefficient.
2. The adapted Cinelli-Hazlett (CH-adapted) fits the short logit model, computes  $\widehat{\text{AME}}_{\text{short}}$  and  $\hat{s}_{\text{short}}$ , and replaces OLS residuals with logit standardised residuals  $\tilde{Y}_i = (Y_i - \hat{\mu}_i) / \sqrt{\hat{\mu}_i(1 - \hat{\mu}_i)}$  when computing  $R_{DZ}^2$  and  $R_{YZ}^2$ . These partial  $R^2$  values and the logit AME are then plugged into (5.2),

$$\widehat{\text{AME}}_{\text{adj}}^{\text{CH-adapted}} = \widehat{\text{AME}}_{\text{short}} - \text{sign}(\widehat{\text{AME}}_{\text{short}}) \sqrt{\frac{\hat{\rho}_{Y,k}^2 \cdot \hat{\rho}_{D,k}^2}{1 - \hat{\rho}_{D,k}^2}} \cdot \hat{s}_{\text{short}} \cdot \sqrt{df}. \quad (5.3)$$

3. The synthetic confounder.

For this experiment, we use DGP (5.1) with parameter values  $\alpha = -0.5$ ,  $\gamma_1 = 0.3$ ,  $\gamma_2 = 0.2$ . The key regressor takes values in  $\beta_D \in \{0, 0.5\}$ . For each value of  $\beta_D$  we consider three levels of confounding strength,  $\delta_Z \in \{0.2, 0.5, 0.8\}$  with corresponding  $\rho_D^{\text{true}} \in \{0.3, 0.4, 0.6\}$  respectively, so that stronger outcome confounding is paired with stronger treatment selection. We generated 10,000 samples of 5,000 observations. For each sample we computed the adjusted AMEs and we report the mean of the absolute errors,  $|\widehat{\text{AME}}_{\text{adj}} - \text{AME}_{\text{true}}|$  (where  $\text{AME}_{\text{true}}$  is computed from the correctly specified logit on a simulated sample of  $n = 500,000$  draws).

The results are presented in Tables 4 and 5. When confounding is weak ( $\delta_Z = 0.2$ ) the nonlinear remainder is small and the CH-OLS provides a good approximation. Tables 4 reports the ratio of Cinelli-Hazlett MAE to Synthetic MAE, which equals 0.94 when  $\beta = 0$  and 1.03 when  $\beta = 0.50$ . As we introduce confounding ( $\delta_Z$  increases) the ratio also

**Table 4:** Adjusted AME accuracy by confounding strength ( $n = 5,000, 10,000$  simulations)

| Panel              | $(\delta_Z, \rho_D)$ | MAE    |            |           | Ratio / Synthetic |             |
|--------------------|----------------------|--------|------------|-----------|-------------------|-------------|
|                    |                      | CH-OLS | CH-adapted | Synthetic | CH-OLS            | CH-adapted  |
| A: $\beta_D = 0$   | (0.2,0.3)            | 0.0115 | 0.0114     | 0.0121    | 0.94              | 0.94        |
|                    | (0.5,0.4)            | 0.0141 | 0.0145     | 0.0129    | 1.09              | 1.12        |
|                    | (0.8,0.6)            | 0.0479 | 0.0547     | 0.0151    | <b>3.17</b>       | <b>3.63</b> |
| B: $\beta_D = 0.5$ | (0.2,0.3)            | 0.0121 | 0.0113     | 0.0117    | 1.03              | 0.96        |
|                    | (0.5,0.4)            | 0.0177 | 0.0170     | 0.0123    | 1.45              | 1.38        |
|                    | (0.8,0.6)            | 0.0651 | 0.0723     | 0.0143    | <b>4.54</b>       | <b>5.04</b> |

Notes: MAE = mean|adjusted AME – true AME|. CH-OLS applies the Cinelli and Hazlett (2020) formula to the short OLS (LPM) model and OLS residuals. CH-adapted applies the same formula to logit standardised residuals and the logit AME. Synthetic re-estimates the full logit at the benchmark  $(\hat{\rho}_{D,Z}, \hat{\rho}_{Y,Z})$ . Ratio = CH MAE / Synthetic MAE; values above 1 favour the synthetic method.

increases. For instance, when  $\beta = 0.5$  it rises from 1.03 at  $\delta_Z = 0.2$  to 1.45 at  $\delta_Z = 0.5$  and 4.54 at  $\delta_Z = 0.8$ . Similar increases are seen when  $\beta_D = 0$ . Table 5 shows that when the Cinelli–Hazlett bias grows roughly sixteen-fold as we increase confounding four-fold from  $\delta_Z = 0.2$  to  $\delta_Z = 0.8$ . This is consistent with the sixteen-fold increase implied by the nonlinear OVB formula.

**Table 5:** Signed bias by confounding strength ( $n = 5,000, 10,000$  simulations)

| Panel              | $(\delta_Z, \rho_D)$ | CH-OLS | CH-adapted | Synthetic |
|--------------------|----------------------|--------|------------|-----------|
| A: $\beta_D = 0$   | (0.2,0.3)            | 0.0027 | 0.0027     | 0.0003    |
|                    | (0.5,0.4)            | 0.0091 | 0.0099     | 0.0008    |
|                    | (0.8,0.6)            | 0.0479 | 0.0547     | 0.0012    |
| B: $\beta_D = 0.5$ | (0.2,0.3)            | 0.0038 | 0.0015     | −0.0011   |
|                    | (0.5,0.4)            | 0.0153 | 0.0146     | −0.0000   |
|                    | (0.8,0.6)            | 0.0651 | 0.0723     | 0.0021    |

Notes: Bias = mean(adjusted AME – true AME); positive values indicate overcorrection, negative values indicate undercorrection. CH-OLS and CH-adapted bias grows with  $\delta_Z$  in both panels, consistent with the  $O(\delta_0^2)$  nonlinear remainder term. The synthetic method’s bias remains close to zero throughout, including in Panel A where the true treatment effect is zero.

Table 5 reports the bias underlying the MAE figures in Table 4. Both Cinelli–Hazlett variants under-correct as  $\delta_Z$  grows, in both the null panel ( $\beta_D = 0$ ) and the positive-effect panel ( $\beta_D = 0.5$ ). This confirms that the MAE differences in Table 4 are driven by bias rather than increased variance, and is consistent with the  $O(\delta_0^2)$  nonlinear remainder term identified in Section 2. That is, the CH correction, being a first-order approximation, fails to absorb the growing second-order component of the bias. Finally, the standard

and adapted Cinelli–Hazlett approaches perform similarly throughout, so that replacing OLS residuals with logit standardised residuals does not improve accuracy.

## 6 Empirical Illustration: Cognitive Skills and Labour Supply in the 1970 British Cohort Study

In recent decades, a large body of research has highlighted the importance of cognitive skills for economic and social outcomes, including educational attainment, employment, earnings, occupational status, health, and social mobility (Heckman et al., 2006; Hanushek and Woessmann, 2008<sup>9</sup>). This research has motivated substantial interest in understanding how cognitive skills develop over the life course and the extent to which cognitive abilities acquired during childhood have lasting effects on outcomes in adulthood. The latter matters because a substantial share of cognitive ability appears to be established early in life, with many cognitive skills exhibiting considerable persistence from late childhood onwards, even after discounting the effect of noncognitive skills (Heckman et al., 2013; Cunha and Heckman, 2007)

Estimating the causal effect of cognitive skills on labour market outcomes is, however, challenging because cognitive ability is endogenous to many of the outcomes researchers seek to explain (such as employment experiences, occupational environments, and educational investments). To address these concerns, researchers have frequently relied on longitudinal cohort studies such as the British Cohort Study (BCS70), which measure skills during childhood or adolescence, well before adulthood outcomes of interest are realised (Del Bono et al., 2024; Golsteyn et al., 2014; Castillo et al., 2019; Angerer et al., 2023; Denny and Doyle, 2008; Castillo et al., 2011; Deary et al., 2000; Schoon et al., 2010; Gill and Prowse, 2024; Fé and Jahangir, 2026). Because cognitive assessments are collected prior to most adult educational, occupational, and family choices, they provide

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<sup>9</sup>Several mechanisms may explain these long-run associations. Higher cognitive skills facilitate learning, the acquisition of greater educational attainment and advanced skills. Cognitive skills may also directly enhance productivity by improving individuals’ ability to process information, solve problems, and adapt to changing environments. Together, these mechanisms may generate positive returns in the labour market through higher employment probabilities, greater earnings, and access to occupations of higher quality.

exogenous measures of skills and reduce concerns regarding biases due to reverse causality and omitted variables. However, they do not fully eliminate the risk, making sensitivity analyses an important step in the research process.

Here we study the robustness of the association between childhood cognitive skills and labour market outcomes decades later, once a large set of predetermined observable confounders is taken into account. The study uses the BCS70, which follows everyone born in Great Britain in a single week of April 1970 (approximately 17,000 births) with sweeps which, at the time of writing, cover ages 5, 10, 16, 26, 30, 34, 38, 42, 46, and 51. For this illustration, the paper focuses on labour market outcomes at the age 46 wave. Specifically, the analysis uses binary indicators of employment and managerial jobs, which are proxies of labour force participation and the quality of jobs. The specification of the short model is,

$$\mathbb{E}(Y_i \mid D_i, \mathbf{X}_i) = \Lambda(\alpha + \beta D_i + \mathbf{X}_i' \boldsymbol{\gamma}), \quad (6.1)$$

where  $\Lambda(\cdot)$  is the logistic CDF. In the absence of a sensitivity analysis, a researcher would be implicitly assuming Conditional Independence, by which

$$\mathbb{E}(Y_i \mid D_i, \mathbf{X}_i, Z_i) = \mathbb{E}(Y_i \mid D_i, \mathbf{X}_i) \quad \text{for all } Z_i, \quad (6.2)$$

that is, conditional on the observed covariates  $\mathbf{X}_i$ , no unobserved variable  $Z_i$  exists that is associated with cognitive skills  $D_i$  and with the outcome  $Y_i$ . Equation (6.2) is equivalent to assuming that any such  $Z_i$  satisfies  $\rho_D = \text{Cor}(Z_i, \tilde{D}_i) = 0$  or  $\rho_Y = \text{Cor}(Z_i, \tilde{Y}_i) = 0$ , where  $\tilde{D}_i$  and  $\tilde{Y}_i$  are the residualised treatment and outcome defined above.

The key treatment variable,  $D$ , is an index of cognitive skills assessed at age 10<sup>10</sup>. Cognitive ability at age 10 was measured via the Friendly Maths Test, consisting of 51 numerical reasoning problems, together with four subtests of the British Ability Scales<sup>11</sup>. Because test administration spanned several months in 1980, each component score was

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<sup>10</sup>The full description of the measure and its construction is given in Fé and Jahangir (2026)

<sup>11</sup>The BAS included the following tasks: Word Definitions (vocabulary and verbal comprehension), Word Similarities (abstract verbal reasoning), Digit Recall (short-term and working memory), and Progressive Matrices (nonverbal pattern recognition, analogous to Raven's Progressive Matrices)

first age-standardised within two-month (bimester) bins defined by the child’s exact age at assessment. As in (Gill and Prowse, 2024; Fé and Jahangir, 2026), the five age-standardised scores were then combined via principal component analysis, retaining the first principal component as a single summary measure of cognitive skill.

The vector  $\mathbf{X}$  includes a rich collection of regressors. Perinatal controls include the mother’s age at delivery (and its square), region of birth, maternal smoking during pregnancy, gestational abnormality, infant sex, birth weight, parity, and prior pregnancy losses. Age-5 controls include breastfeeding, inpatient hospital episodes, sensory or speech difficulties, whether the child was read to, parental qualifications and employment status, ethnicity, and number of siblings. Age-10 controls include further inpatient episodes, parental unemployment and benefit receipt, household income band, housing tenure, number of rooms, parental smoking, sensory or speech difficulties, standardised height and weight, and teacher-reported disengagement.

The vector  $\mathbf{X}$  further includes teacher-reported measures of the child’s levels of developmental behaviour at age 10. This assessment is based on a multiple item questionnaire that captures various aspects of the child’s character. From the questionnaire we keep a raw measure of perseverance (Fé and Jahangir, 2026), alongside six principal components summarising developmental behaviours orthogonal to perseverance<sup>12</sup>. The first component “Emotional Sensitivity and Social Dysfunction”, contrasts aggressive social behaviours such as teasing and bullying against emotional vulnerability indicators like nervousness and clumsiness. The second component, “Anxiety and Fine Motor Difficulties”, combines anxiety-related behaviours with fine motor skill challenges, potentially reflecting developmental coordination difficulties. The remaining components capture more specific dimensions. Component 3 relates to incontinence and physical coordination problems. Component 4 reflects a contrast between impulsivity and social withdrawal. Component 5 combines motor skill deficits with stereotypic behaviours such as tics and repetitive vocalisations, possibly associated with neurodevelopmental conditions. Finally, Component 6 emphasises severe motor coordination difficulties.

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<sup>12</sup>The construction of these components is discussed in Fé and Jahangir (2026).

**Table 6:** Average marginal effects on full-time employment and managerial status at age 46 (significant effects only).

|                                       | Full-time employment |          | Manager/Supervisor |          |
|---------------------------------------|----------------------|----------|--------------------|----------|
|                                       | AME                  | SE       | AME                | SE       |
| Cognitive skills                      | 0.0267***            | (0.0062) | 0.0291***          | (0.0083) |
| Female                                | −0.4061***           | (0.0151) | −0.1466***         | (0.0225) |
| Father unemployed at age 10           | 0.1186***            | (0.0375) | 0.1515**           | (0.0738) |
| Behavioural PC6                       | 0.0175**             | (0.0086) | —                  | —        |
| Behavioural PC1                       | —                    | —        | 0.0199***          | (0.0058) |
| Region 5 <sup>a</sup>                 | −0.0901***           | (0.0344) | —                  | —        |
| Region 6                              | −0.1253**            | (0.0504) | —                  | —        |
| Region 7                              | —                    | —        | 0.0903*            | (0.0468) |
| Mother has no qualifications          | 0.0538*              | (0.0318) | —                  | —        |
| Mother stays home at age 10           | −0.0337*             | (0.0203) | —                  | —        |
| Mother stayed in education beyond SLA | —                    | —        | 0.0707**           | (0.0283) |
| Child was read to at age 5            | —                    | —        | 0.0599**           | (0.0255) |
| Non-white ethnicity                   | —                    | —        | 0.1058**           | (0.0429) |
| Standardised height at age 10         | —                    | —        | 0.0464***          | (0.0150) |
| Standardised weight at age 10         | —                    | —        | −0.0323**          | (0.0149) |
| Mother works full-time at age 5       | —                    | —        | −0.0589*           | (0.0313) |
| Income band 1 (lowest)                | —                    | —        | −0.1064*           | (0.0600) |
| Housing tenure category 2             | —                    | —        | −0.3822*           | (0.1990) |
| Housing tenure category 3             | —                    | —        | −0.3536*           | (0.1940) |
| Housing tenure category 4             | —                    | —        | −0.3370**          | (0.1532) |
| Housing tenure category 5             | −0.5166**            | (0.2621) | —                  | —        |
| Observations                          | 3,584                |          | 2,420              |          |
| Pseudo $R^2$                          | 0.181                |          | 0.065              |          |

<sup>a</sup> Region indicators are coded relative to the omitted reference category (Region 10).

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 6 reports AMEs for both outcomes, restricted to covariates significant at the 10% level or better. Cognitive skills predict both outcomes strongly and significantly. A one-unit increase in the index of cognitive skills raises the probability of full-time employment by 2.7 percentage points and the probability of holding a managerial or supervisory position by 2.9 percentage points.

**Table 7:** Fragility thresholds by outcome and  $\rho_Y$ , BCS70 age-46 labour supply.

| $\rho_Y$ | Full-time employment |            | Manager/Supervisor |            |
|----------|----------------------|------------|--------------------|------------|
|          | $\rho_D^*$           | $\rho_D^+$ | $\rho_D^*$         | $\rho_D^+$ |
| -0.5     | -0.104               | -0.198     | -0.126             | -0.248     |
| -0.4     | -0.115               | -0.223     | -0.135             | -0.278     |
| -0.3     | -0.139               | -0.279     | -0.157             | -0.349     |
| -0.2     | -0.201               | -0.442     | -0.215             | -0.603     |
| -0.1     | -0.449               | robust     | -0.484             | robust     |
| 0.0      | 0.871                | robust     | 0.621              | 0.931      |
| 0.1      | 0.333                | 0.515      | 0.217              | 0.449      |
| 0.2      | 0.202                | 0.328      | 0.123              | 0.277      |
| 0.3      | 0.150                | 0.241      | 0.086              | 0.197      |
| 0.4      | 0.125                | 0.193      | 0.070              | 0.153      |
| 0.5      | 0.112                | 0.166      | 0.061              | 0.126      |

$\rho_D^*$  is the explain-away threshold;  $\rho_D^+$  is the sign-change threshold. “robust” indicates that no threshold is reached within the feasible range. Short-model AMEs: 0.0233 (SE 0.0054) for full-time employment; 0.0266 (SE 0.0075) for managerial/supervisory status.

## 6.1 Sensitivity Analysis

We applied the synthetic confounder method to the short model. The fragility thresholds<sup>13</sup> are presented in Table 7. At  $\rho_Y = 0$  the thresholds sits near the boundary of the feasibility region in Proposition 3.1 (0.871 for full-time employment, 0.621 for managerial status) as we expected, since a confounder that does not move the outcome at all cannot bias the estimate. As we move away from zero, however, the threshold falls quickly. By  $\rho_Y = \pm 0.1$  (a confounder explaining a 1% of the outcome residual’s variance) the explain-away threshold has already dropped to 0.333 for full-time employment and 0.217

<sup>13</sup>Recall from Section 3 that we may read the results as follows. A  $\rho_Y = 0.1$  is a correlation, and the corresponding variance share is  $\rho_Y^2$ , so  $\rho_Y = 0.1$  corresponds to a confounder explaining only 1% of the variance in the outcome residual by outcome

for managerial status.

The table also shows that fragility depends on the direction of the hypothetical confounder. For  $\rho_Y < 0$  (a confounder that would suppress the outcome net of cognition) full-time employment’s threshold is smaller in magnitude than managerial status’ at every row (e.g.  $-0.104$  versus  $-0.126$  at  $\rho_Y = -0.5$ ), making employment the more fragile outcome in this direction. For  $\rho_Y > 0$ , managerial status’ threshold is smaller at every row ( $0.217$  versus  $0.333$  at  $\rho_Y = 0.1$ ;  $0.123$  versus  $0.202$  at  $\rho_Y = 0.2$ ; and so on), making managerial status the more fragile outcome. This second direction is also the more plausible one to worry about in this setting because the candidate omitted confounders will include family socioeconomic advantage, parental ability, or early-childhood investment, which are factor expected to raise both childhood cognitive skill and adult labour-market success simultaneously.

In general, the explain away threshold  $\rho_D^*$  never reaches “robust” status at any  $\rho_Y$  in the range considered for either outcome. Thus, we conclude that a sufficiently strong confounder always exists that would drive cognitive skills’ association with labour-market status to statistical insignificance. For full-time employment, sign reversal is infeasible within the scanned grid at both  $\rho_Y = -0.1$  and  $\rho_Y = 0.0$ , while for managerial status, only  $\rho_Y = -0.1$  shows this.

As explained in Section 4.1, it is possible to tie the choice of  $\rho_Y$  to the current application at hand by asking how much would the estimate shift if a confounder as strong as a specific observed variable had been omitted. In principle, the choice of benchmarks can be informed by the regression results in Table 6. The largest marginal effect in either model belongs to sex, with women being significantly less likely to be in full-time employment or to hold a managerial position than men. Father’s unemployment at age 10 is also a strong significant predictor of both outcomes. Therefore, on the basis of their strong predictive power, the benchmarking analysis could rely on sex, father’s unemployment status at age 10 and whether the mother stayed at home at age 10 as benchmarks. However, our main concern is the existence of confounder which are not orthogonal to treatment (those which are do not bias the estimates of the treatment effect).

**Table 8:** Benchmark analysis: strongest observed predictors, BCS70 age-46 labour supply.

*FT Employment*

| Variable          | $\hat{\rho}_D$ | $\hat{R}_D^2$ | $\hat{\rho}_Y$ | $\hat{R}_Y^2$ | ME     | $t$  | Bias    |
|-------------------|----------------|---------------|----------------|---------------|--------|------|---------|
| Sex               | -0.074         | 0.55%         | -0.399         | 15.92%        | 0.0138 | 2.76 | 0.0095  |
| Mother stay home  | -0.009         | 0.01%         | -0.032         | 0.10%         | 0.0230 | 4.30 | 0.0003  |
| Father unemployed | -0.012         | 0.01%         | 0.049          | 0.24%         | 0.0239 | 4.46 | -0.0006 |
| PC1               | 0.168          | 2.82%         | 0.004          | 0.00%         | 0.0228 | 4.21 | 0.0005  |
| Height at 10      | 0.134          | 1.80%         | 0.026          | 0.07%         | 0.0221 | 4.09 | 0.0012  |
| Weight at 10      | 0.082          | 0.67%         | -0.030         | 0.09%         | 0.0238 | 4.44 | -0.0005 |

*Manager*

| Variable          | $\hat{\rho}_D$ | $\hat{R}_D^2$ | $\hat{\rho}_Y$ | $\hat{R}_Y^2$ | ME     | $t$  | Bias    |
|-------------------|----------------|---------------|----------------|---------------|--------|------|---------|
| Sex               | -0.095         | 0.90%         | -0.130         | 1.69%         | 0.0236 | 3.14 | 0.0030  |
| Mother stay home  | -0.006         | 0.00%         | -0.023         | 0.05%         | 0.0266 | 3.52 | -0.0000 |
| Father unemployed | -0.008         | 0.01%         | 0.038          | 0.14%         | 0.0263 | 3.49 | 0.0003  |
| PC1               | 0.161          | 2.59%         | 0.069          | 0.48%         | 0.0202 | 2.63 | 0.0064  |
| Height at 10      | 0.115          | 1.32%         | 0.063          | 0.40%         | 0.0221 | 2.91 | 0.0044  |
| Weight at 10      | 0.105          | 1.10%         | -0.045         | 0.20%         | 0.0276 | 3.63 | -0.0010 |

ME is the adjusted AME evaluated at  $(\hat{\rho}_D, \hat{\rho}_Y)$ . Bias = ME<sub>short</sub> - ME<sub>adjusted</sub>.  $\hat{R}_D^2 = \hat{\rho}_D^2$  and  $\hat{R}_Y^2 = \hat{\rho}_Y^2$  are the implied shares of residual (post-control) treatment and outcome variance respectively.

Table 8, which evaluates the fragility of the results against the selected benchmarks, reveals that none of the previous predictors are strongly associated with cognitive skills. Specifically, for sex, father’s unemployment and mother staying home at age 10,  $|\hat{\rho}_D| \leq 0.095$ . For the latter two benchmarks, this falls below the corresponding explain-away thresholds, but because of the weak association with the treatment, the implied bias should be small (and this is confirmed by the results).

Given this, the paper considers a second set of benchmarks based on the largest partial correlations with cognitive skills, presented in Table 14 in Appendix B. In particular, the selected benchmarks are age standardised weight and height and the first behavioural principal component (PC1), capturing emotional sensitivity and social dysfunction. Table 8 shows how these predictors have, in general, strong associations with the residual variation in treatment (in the 1% - 2.8% range). Note that although these values are modest in absolute terms, this range is typical for partial  $R^2$  after a rich control set.

Of every benchmark considered, PC1 comes closest to its explain-away threshold  $\rho_D^*$ , that is, its treatment correlation  $\hat{\rho}_D$  gets close to the point at which a confounder of that strength would render the result statistically insignificant. For full-time employment, PC1 has coordinates  $(\hat{\rho}_D, \hat{\rho}_Y) = (0.168, 0.004)$ . The nearest row of Table 7 ( $\rho_Y = 0.0$ ) has an explain away threshold of  $\rho_D^* = 0.871$ , leaving a wide margin of about 80% ( $0.168/0.871 \approx 0.19$ ). For managerial status, however, the coordinates are  $(0.161, 0.069)$ , and the nearest threshold ( $\rho_D^* = 0.217$  at  $\rho_Y = 0.1$ ) leaves a margin of about 25%.

Overall, confounding as strong as the strongest benchmarks considered do not explain away the association between cognitive skills and managerial status. However, we understand that a confounder only modestly stronger than the PC1 would be sufficient to overturn the conclusion. The association with full time employment remains robust to confounding at the strength of any of the benchmarks considered.

## 7 Conclusion

This paper has developed a sensitivity analysis framework for likelihood-based nonlinear models, inspired by Cinelli and Hazlett (2020)’s approach for linear regression. The paper first discusses how the omitted variable bias of a likelihood-based estimator contains a non-vanishing remainder term that grows quadratically with the strength of the confounder and depends on the curvature of the model’s nonlinearity. The implication is that sensitivity methods designed for linear models can be applied directly to nonlinear models, but only as a first-order approximation that performs well under weak confounding. Under moderate to strong confounding, such methods struggle to recover the true magnitude of the bias.

In view of this result, the paper proposes a simulation-based method to construct synthetic confounders with exact, pre-specified partial correlations with the residualised treatment and outcome to recover the adjusted AME by re-estimating the long model. Using this method, the paper develops a fragility framework that allows researchers to charts the strength of confounding required to eliminate the statistical significance of the estimated effect or to reverse its sign.

Because the partial correlations of unobserved confounders are not known in practice, the paper also develops a benchmarking procedure, following Cinelli and Hazlett (2020), that grounds the sensitivity analysis in the specific empirical application. The procedure asks how the result would change if an omitted confounder were as strongly correlated with treatment and outcome as a specific observed covariate. The benchmark estimates can then be compared against the fragility thresholds to assess how plausible it is that confounding of the relevant magnitude could overturn the conclusion.

The exposition has assumed, for convenience, that the confounder enters the outcome equation homogeneously, without interacting with the treatment. However, homogeneity is not required since the synthetic confounder is built directly from the partial correlations  $(\rho_D, \rho_Y)$  rather than from a structural coefficient, and treatment-effect heterogeneity can be accommodated by re-estimating the long model with an added interaction term, as discussed in Section ??.

A Monte Carlo simulation evaluates the performance of the proposed procedures, showing that direct application of linear-model sensitivity methods to nonlinear models is only a first-order approximation whose accuracy deteriorates as confounding strengthens, while the synthetic confounder method remains accurate throughout, including under strong confounding.

The method is applied to assess the robustness of cognitive skills measured at age 10 as a predictor of labour-market outcomes in adulthood. Using data from the 1970 British Cohort Study, the paper reports estimates of short models of labour force participation (full-time employment) and job quality (managerial or supervisory status) that include a composite measure of cognitive skills alongside a rich set of pre-treatment covariates, including parental background and parent- and teacher-reported measures of developmental behaviour. The concern motivating the analysis is that, despite the richness of this control set, residual sources of confounding arising from parental background, environmental factors, and similar unobserved influences may remain. Such confounders are likely to be positively associated with cognitive skills and adult labour-market outcomes, which would inflate the estimated effect of cognitive skill. We find that cognitive skills remain a robust predictor of employment participation, but are a comparatively fragile predictor of job quality. In particular, a benchmark only modestly stronger than the strongest observed covariate would be sufficient to overturn the latter conclusion.

The framework presented here has a number of limitations that point to directions for future research. In a recent paper, Diegert et al. (2025) develop axiomatic results for sensitivity frameworks. Extending that axiomatic treatment to the simulation-based setting is a natural but non-trivial next step, and is left for future research. As in Cinelli and Hazlett (2020), the partial correlations are fixed to levels decided by the researcher, and one might wish to allow for uncertainty in these choices, allowing for sampling variability around the sensitivity parameters themselves. The framework has also been developed with a single confounder in mind. In practice, multiple confounders may operate simultaneously. The single synthetic confounder can, however, be interpreted as a summary of their combined influence. Nonetheless, how to handle multiple unobservable quanti-

ties whose interactions with treatment and outcome are complex and distinct from one another remains an open question. Despite these limitations, the synthetic confounder framework offers applied researchers a flexible tool for assessing the robustness of causal conclusions in the nonlinear models that are routine in empirical research.

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# A Appendix

## A.1 Regularity Conditions.

Write  $\boldsymbol{\xi} = (\boldsymbol{\theta}', \delta)' \in \mathbb{R}^{p+3}$  for the long-model parameter, where  $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \beta)' \in \mathbb{R}^{p+2}$  is the short-model parameter,  $\boldsymbol{\alpha} \in \mathbb{R}^{p+1}$  collects coefficients on  $A_i = (1, \mathbf{X}_i')' \in \mathbb{R}^{p+1}$ ,  $\mathbf{X}_i \in \mathbb{R}^p$ ,  $\beta \in \mathbb{R}$  is the coefficient on  $D_i$ , and  $\delta \in \mathbb{R}$  is the coefficient on the omitted variable  $Z_i$ . The true parameter is  $\boldsymbol{\xi}_0 = (\boldsymbol{\theta}_0', \delta_0)'$ .

The long-model score is

$$\mathbf{s}_i(\boldsymbol{\xi}) := \frac{\partial \ell_i(\boldsymbol{\xi})}{\partial \boldsymbol{\xi}} = \begin{pmatrix} \mathbf{s}_{i,\theta}(\boldsymbol{\theta}, \delta) \\ s_{i,\delta}(\boldsymbol{\theta}, \delta) \end{pmatrix} \in \mathbb{R}^{p+3},$$

where  $\mathbf{s}_{i,\theta}(\boldsymbol{\theta}, \delta) := \partial \ell_i(\boldsymbol{\theta}, \delta) / \partial \boldsymbol{\theta}$  and  $s_{i,\delta}(\boldsymbol{\theta}, \delta) := \partial \ell_i(\boldsymbol{\theta}, \delta) / \partial \delta$ . The corresponding long-model Hessian is

$$\mathbf{H}_i(\boldsymbol{\xi}) := \frac{\partial^2 \ell_i(\boldsymbol{\xi})}{\partial \boldsymbol{\xi} \partial \boldsymbol{\xi}'} = \begin{pmatrix} \mathbf{H}_{i,\theta\theta}(\boldsymbol{\xi}) & \mathbf{H}_{i,\theta\delta}(\boldsymbol{\xi}) \\ \mathbf{H}_{i,\delta\theta}(\boldsymbol{\xi}) & H_{i,\delta\delta}(\boldsymbol{\xi}) \end{pmatrix}.$$

The short model is the long model with  $\delta$  restricted to zero, so that its log-likelihood is  $\ell_i^*(\boldsymbol{\theta}) = \ell_i(\boldsymbol{\theta}, 0)$ , its score is  $\mathbf{s}_i^*(\boldsymbol{\theta}) := \partial \ell_i^*(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \in \mathbb{R}^{p+2}$ , and its Hessian is  $\mathbf{H}_i^*(\boldsymbol{\theta})$ . Hence, pointwise,

$$\mathbf{s}_i^*(\boldsymbol{\theta}) = \mathbf{s}_{i,\theta}(\boldsymbol{\theta}, 0).$$

The pseudo-true parameter  $\boldsymbol{\theta}^*$  is the maximiser of the expected short-model log-likelihood,

$$\boldsymbol{\theta}^* := \arg \max_{\boldsymbol{\theta}} \mathbb{E}[\ell_i^*(\boldsymbol{\theta})].$$

The following standard conditions are assumed to hold throughout.

**R1.** The long-model parameter space  $\Xi \subset \mathbb{R}^{p+3}$  is compact, and  $\boldsymbol{\xi}_0 = (\boldsymbol{\theta}_0', \delta_0)'$  is an interior point of  $\Xi$ . The short-model parameter  $\boldsymbol{\theta}$  lies in  $\Theta := \{\boldsymbol{\theta} : (\boldsymbol{\theta}', 0)' \in \Xi\}$ , with  $\boldsymbol{\theta}_0$  and  $\boldsymbol{\theta}^*$  interior points of  $\Theta$ .

**R2.** The long-model log-likelihood  $\ell_i(\boldsymbol{\xi})$  is three times continuously differentiable in  $\boldsymbol{\xi}$

on  $\Xi$ , almost surely (a.s.), and the short-model log-likelihood  $\ell_i^*(\boldsymbol{\theta})$  is three times continuously differentiable in  $\boldsymbol{\theta}$ , a.s.

**R3.** The regressors  $W_i$  satisfy a moment condition sufficient for dominated convergence of the score, Hessian, and third derivatives of both the long and short log-likelihoods under the model family.<sup>14</sup>

**R4.** There exist square-integrable functions  $g_1(Y_i, W_i)$  and  $g_1^*(Y_i, W_i)$  with  $\mathbb{E}[g_1^2], \mathbb{E}[(g_1^*)^2] < \infty$  such that

$$\|\mathbf{s}_i(\boldsymbol{\xi})\| \leq g_1(Y_i, W_i) \quad \text{for all } \boldsymbol{\xi} \in \Xi,$$

and

$$\|\mathbf{s}_i^*(\boldsymbol{\theta})\| \leq g_1^*(Y_i, W_i) \quad \text{for all } \boldsymbol{\theta} \in \Theta.$$

**R5.** There exist integrable functions  $g_2(Y_i, W_i)$ ,  $g_2^*(Y_i, W_i)$ ,  $g_3(Y_i, W_i)$ , and  $g_3^*(Y_i, W_i)$  with finite expectations such that

$$\|\mathbf{H}_i(\boldsymbol{\xi})\| \leq g_2(Y_i, W_i), \quad \|\mathbf{H}_i^*(\boldsymbol{\theta})\| \leq g_2^*(Y_i, W_i),$$

for all  $\boldsymbol{\xi} \in \Xi$  and  $\boldsymbol{\theta} \in \Theta$ , respectively, and

$$\left\| \frac{\partial^3 \ell_i(\boldsymbol{\xi})}{\partial \xi_j \partial \xi_k \partial \xi_l} \right\| \leq g_3(Y_i, W_i)$$

for all  $\boldsymbol{\xi} \in \Xi$  and all indices  $j, k, l$ . The corresponding third derivatives of  $\ell_i^*(\boldsymbol{\theta})$  are bounded by  $g_3^*(Y_i, W_i)$  uniformly over  $\boldsymbol{\theta} \in \Theta$ . Here  $\xi_j$  denotes any component of  $\boldsymbol{\xi} = (\boldsymbol{\theta}', \delta)'$ .

**R6.** The population information matrices  $\mathcal{I}^*(\boldsymbol{\theta}_0)$  and  $\mathcal{I}(\boldsymbol{\xi}_0)$  are both finite and positive definite, where

$$\mathcal{I}^*(\boldsymbol{\theta}) := -\mathbb{E}[\mathbf{H}_i^*(\boldsymbol{\theta})]$$

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<sup>14</sup>The required strength varies by model. In bounded-variance models (logit, cloglog)  $\mathbb{E}[\|W_i\|^3] < \infty$  suffices. In unbounded-variance models (Poisson),  $\mathbb{E}[e^{c\|W_i\|}] < \infty$  for some  $c > 0$  is required.

and

$$\mathcal{I}(\boldsymbol{\xi}) := -\mathbb{E}[\mathbf{H}_i(\boldsymbol{\xi})]$$

denote the short-model information matrix and the full long-model information matrix, respectively.

**R7.** There exists an integrable function  $g_{12}(Y_i, W_i)$  with  $\mathbb{E}[g_{12}] < \infty$  such that

$$\|\mathbf{s}_i^*(\boldsymbol{\theta})\| |s_{i,\delta}(\boldsymbol{\theta}, \delta)|^2 \leq g_{12}(Y_i, W_i) \quad \text{and} \quad \|\mathbf{s}_i^*(\boldsymbol{\theta})\| |s_{i,\delta\delta}(\boldsymbol{\theta}, \delta)| \leq g_{12}(Y_i, W_i)$$

for all  $(\boldsymbol{\theta}, \delta) \in \Xi$ , where  $s_{i,\delta\delta} := \partial^2 \ell_i(\boldsymbol{\theta}, \delta) / \partial \delta^2$ . Furthermore, there exists an integrable function  $g_{21}(Y_i, W_i)$  with  $\mathbb{E}[g_{21}] < \infty$  such that

$$\|\mathbf{H}_i^*(\boldsymbol{\theta})\| |s_{i,\delta}(\boldsymbol{\theta}, \delta)| \leq g_{21}(Y_i, W_i)$$

for all  $(\boldsymbol{\theta}, \delta) \in \Xi$ .

The distributions  $\{P_\delta : (\boldsymbol{\theta}, \delta) \in \Xi\}$  admit a common  $\sigma$ -finite dominating measure  $\lambda$ , with densities  $p(y, w; \delta)$  with respect to  $\lambda$ , and the envelope conditions in **R4**, **R5**, and **R7** are understood to provide integrable dominating functions for the corresponding density-weighted integrands with respect to  $\lambda$ .

Assumptions **R1**–**R6** are standard in the literature (e.g. White, 1982). Condition **R4** provides the score envelopes for both the long and short models and justifies differentiation under the expectation when deriving the properties of their population scores. The Hessian envelopes in **R5** justify exchanging differentiation and expectation for the corresponding Hessians. In particular, defining the population long-model score by

$$\mathbf{s}(\boldsymbol{\xi}) := \mathbb{E}_\delta[\mathbf{s}_i(\boldsymbol{\xi})],$$

its  $\boldsymbol{\theta}$ -component satisfies

$$\mathbf{s}_\theta(\boldsymbol{\xi}) = \mathbb{E}_\delta[\mathbf{s}_{i,\theta}(\boldsymbol{\xi})],$$

and

$$\frac{\partial \mathbf{s}_\theta(\boldsymbol{\xi})}{\partial \boldsymbol{\theta}'} = \mathbb{E}_\delta[\mathbf{H}_{i,\theta\theta}(\boldsymbol{\xi})] = -\mathcal{I}_{\theta\theta}(\boldsymbol{\xi}),$$

where  $\mathcal{I}_{\theta\theta}(\boldsymbol{\xi})$  is the  $\boldsymbol{\theta}$ - $\boldsymbol{\theta}$  block of the full long-model information matrix  $\mathcal{I}(\boldsymbol{\xi})$ .

Under these conditions, together with the usual identification condition and a uniform law of large numbers, consistency of  $\hat{\boldsymbol{\theta}}^*$  to  $\boldsymbol{\theta}^*$  follows from **R1**–**R6** and a uniform law of large numbers (White, 1982).

Condition **R7** has several purposes. First, it is required to establish the order of the nonlinear omitted variable bias. Second, **R7** addresses a technical requirement arising from the fact that the distribution of the data changes with the confounding strength  $\delta$ . As a result, integrals defining expected values of scores and their derivatives are taken with respect to  $dP_\delta$ , which varies with  $\delta$ . The dominated convergence theorem (which is used to establish the omitted variable bias), however, requires all integrals to be expressed relative to a single fixed measure. A common dominating measure  $\lambda$  (Lebesgue measure for continuous outcomes, counting measure for discrete outcomes) provides this. Thus, writing  $dP_\delta = p(y, w; \delta) d\lambda$  absorbs the dependence on  $\delta$  into the integrand, so that all integrals are taken with respect to the fixed  $d\lambda$ . The envelope bounds in **R4**–**R7** are then required to hold for these full integrands, including the density factor  $p(y, w; \delta)$ .

## A.2 Supporting Results

Before introducing the main results and their proofs, we collect a number of well known results that are used repeatedly. Under a correctly specified model, the expected score at the true parameter is zero. This follows from differentiating  $\int f(y, w; \boldsymbol{\xi}_0) d\lambda(y, w) = 1$  under the integral sign, justified by **R2** and the domination in **R4**:

$$\mathbb{E}[\mathbf{s}_i(\boldsymbol{\xi}_0)] = \mathbf{0}. \tag{A.1}$$

For every observation  $i$  and every  $\boldsymbol{\theta}$ ,

$$\ell_i^*(\boldsymbol{\theta}) = \ell_i(\boldsymbol{\theta}, 0), \tag{A.2}$$

and differentiating both sides with respect to  $\boldsymbol{\theta}$  (**R2**) recovers the score identity from the notation paragraph:

$$\mathbf{s}_i^*(\boldsymbol{\theta}) = \mathbf{s}_{i,\theta}(\boldsymbol{\theta}, 0) \quad \text{pointwise.} \quad (\text{A.3})$$

This has several implications at  $\delta_0 = 0$ . When the true confounding strength is  $\delta_0 = 0$ , the true long-model parameter is  $\boldsymbol{\xi}_0 = (\boldsymbol{\theta}'_0, 0)'$ . Applying (A.1), all components of the long-model score have expectation zero:

$$\mathbb{E}_0[\mathbf{s}_i(\boldsymbol{\xi}_0)] = \mathbf{0}, \quad (\text{A.4})$$

including the  $\boldsymbol{\theta}$ -components:

$$\mathbb{E}_0[\mathbf{s}_{i,\theta}(\boldsymbol{\theta}_0, 0)] = \mathbf{0}, \quad (\text{A.5})$$

and the  $\delta$ -component:

$$\mathbb{E}_0[s_{i,\delta}(\boldsymbol{\theta}_0, 0)] = 0. \quad (\text{A.6})$$

By (A.3), the short-model score inherits this:

$$\mathbb{E}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0)] = \mathbb{E}_0[\mathbf{s}_{i,\theta}(\boldsymbol{\theta}_0, 0)] = \mathbf{0}. \quad (\text{A.7})$$

Since  $\boldsymbol{\theta}^*$  is an interior maximiser of  $\mathbb{E}_{\delta_0}[\ell_i^*(\boldsymbol{\theta})]$  by **R1** and **R2**, the first-order condition holds. Combined with (A.3):

$$\mathbb{E}_{\delta_0}[\mathbf{s}_{i,\theta}(\boldsymbol{\theta}^*, 0)] = \mathbb{E}_{\delta_0}[\mathbf{s}_i^*(\boldsymbol{\theta}^*)] = \mathbf{0}. \quad (\text{A.8})$$

We wish to establish that, in a neighbourhood of  $\delta_0 = 0$ , there exists a smooth function mapping the confounding strength  $\delta_0$  to the local solution of the short-model population score equation near  $\boldsymbol{\theta}_0$ . To do so we apply the Implicit Function Theorem to the short-model population score equation, expanding around the point  $(\boldsymbol{\theta}_0, 0)$  where there is no confounding and the short model is correctly specified (so  $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0$ ). The preliminary results and assumptions above guarantee that the conditions of the IFT are satisfied. This theorem delivers the existence of  $\boldsymbol{\theta}^*(\delta_0)$  as a continuously differentiable function

and an expression for its derivative at  $\delta_0 = 0$ , which characterises the direction in which the pseudo-true parameter departs from the truth. However, to establish the *rate* at which the pseudo-true parameter departs from the truth as confounding is introduced, we need  $\boldsymbol{\theta}^*(\delta_0)$  to be *twice* continuously differentiable. For this, in addition to conditions **R1**–**R6**, we require the cross-moment envelopes in **R7** and a  $C^k$  version of the IFT, which are introduced after the following ancillary result.

**Lemma A.1.** *Under conditions **R1**–**R7**, the function*

$$F(\boldsymbol{\theta}, \delta) := \mathbb{E}_\delta[\mathbf{s}_i^*(\boldsymbol{\theta})]$$

*is twice continuously differentiable on  $\text{int}(\Xi)$ .*

*Proof.* Since  $(\boldsymbol{\theta}'_0, 0)'$  is an interior point of the compact set  $\Xi$  by **R1**, the interior  $\text{int}(\Xi)$  is an open set containing  $(\boldsymbol{\theta}'_0, 0)'$ , and we work on this open set throughout.

By **R2**, the short-model score  $\mathbf{s}_i^*(\boldsymbol{\theta})$  is twice continuously differentiable in  $\boldsymbol{\theta}$ , with derivatives given by the Hessian and the third derivatives of the short-model log-likelihood. Writing  $\boldsymbol{\xi} = (\boldsymbol{\theta}', \delta)'$  and expressing  $F$  against the fixed base measure  $\lambda$  with  $p(y, w; \delta) := p(y, w; \boldsymbol{\theta}_0, \delta)$  denoting the true DGP density,

$$F(\boldsymbol{\theta}, \delta) = \int \mathbf{s}_i^*(\boldsymbol{\theta}) p(y, w; \delta) d\lambda(y, w),$$

where  $\boldsymbol{\theta}$  is the argument of the short-model score (being optimised) and  $\delta$  enters only through the density of the true DGP. By Corollary A.2 (in Appendix A.6), for a generic point  $\boldsymbol{\xi} = (\boldsymbol{\theta}', \delta)' \in \text{int}(\Xi)$ , the short-model Hessian envelope  $g_2^*$  in **R5**, together with the common dominating-measure condition in **R7**, justifies the interchange, giving

$$\frac{\partial F}{\partial \theta_j}(\boldsymbol{\theta}, \delta) = \int \frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta})}{\partial \theta_j} p(y, w; \delta) d\lambda(y, w)$$

for every  $\boldsymbol{\xi}$ .

Continuity is established by showing that the integral of the derivative at  $\boldsymbol{\xi}_n$  converges to the integral of the derivative at  $\boldsymbol{\xi}_0$  for arbitrary  $\boldsymbol{\xi}_n \rightarrow \boldsymbol{\xi}_0$ , and then invoke

the sequential characterisation of continuity (e.g. Theorem 5.1.3 in Bartle and Sherbert, 2011). Therefore, consider an arbitrary sequence  $\boldsymbol{\xi}_n \rightarrow \boldsymbol{\xi}_0$ . By **R2** and the continuity of the density in  $\delta$ ,

$$\frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta}_n)}{\partial \theta_j} p(y, w; \delta_n) \longrightarrow \frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta}_0)}{\partial \theta_j} p(y, w; \delta_0)$$

and the density-weighted envelope conditions in **R5** and **R7** provide an integrable dominating function for the full integrand (including the density factor), so the same envelope that justified the first step also dominates every term in the sequence. The Dominated Convergence Theorem therefore gives

$$\int \frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta}_n)}{\partial \theta_j} p(y, w; \delta_n) d\lambda(y, w) \longrightarrow \int \frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta}_0)}{\partial \theta_j} p(y, w; \delta_0) d\lambda(y, w).$$

Combining the previous results:

$$\frac{\partial F}{\partial \theta_j}(\boldsymbol{\theta}_n, \delta_n) = \int \frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta}_n)}{\partial \theta_j} p(y, w; \delta_n) d\lambda(y, w) \longrightarrow \int \frac{\partial \mathbf{s}_i^*(\boldsymbol{\theta}_0)}{\partial \theta_j} p(y, w; \delta_0) d\lambda(y, w) = \frac{\partial F}{\partial \theta_j}(\boldsymbol{\theta}_0, \delta_0).$$

Since the sequence  $\boldsymbol{\xi}_n \rightarrow \boldsymbol{\xi}_0$  was arbitrary, this establishes continuity of  $\partial F / \partial \theta_j$  at  $\boldsymbol{\xi}_0$  by the sequential characterisation of continuity.

The same two-stage argument applies to each second-order derivative of  $F$ . Specifically,

$$\frac{\partial F}{\partial \delta} = \mathbb{E}_\delta[\mathbf{s}_i^*(\boldsymbol{\theta}) s_{i,\delta}(\boldsymbol{\theta}_0, \delta)], \quad \text{integrable by **R4** and Cauchy–Schwarz} \quad (\text{A.9})$$

$$\frac{\partial^2 F}{\partial \delta^2} = \mathbb{E}_\delta[\mathbf{s}_i^*(\boldsymbol{\theta}) s_{i,\delta\delta}(\boldsymbol{\theta}_0, \delta)] + \mathbb{E}_\delta[\mathbf{s}_i^*(\boldsymbol{\theta}) s_{i,\delta}(\boldsymbol{\theta}_0, \delta)^2], \quad \text{envelope } g_{12}, \text{ **R7**} \quad (\text{A.10})$$

$$\frac{\partial^2 F}{\partial \boldsymbol{\theta}' \partial \delta} = \mathbb{E}_\delta[\mathbf{H}_i^*(\boldsymbol{\theta}) s_{i,\delta}(\boldsymbol{\theta}_0, \delta)]. \quad \text{envelope } g_{21}, \text{ **R7**} \quad (\text{A.11})$$

Thus all first- and second-order partial derivatives of  $F$  exist and are continuous on  $\text{int}(\Xi)$ . Hence  $F \in C^2$ . ■

The previous result allows us to conclude that the expected value of the short-model score is  $C^2$ . This, in turn, enables application of the following result and the proof of the

omitted variable bias.

**Theorem A.1** ( $C^k$  Implicit Function Theorem; Krantz and Parks, 2002, Theorem 3.3.1).

Let  $F : U \times V \rightarrow \mathbb{R}^q$  be a  $C^k$  function,  $k \geq 1$ , where  $U \subseteq \mathbb{R}^q$  and  $V \subseteq \mathbb{R}$  are open sets. Suppose there exists a point  $(\mathbf{x}_0, y_0) \in U \times V$  such that  $F(\mathbf{x}_0, y_0) = \mathbf{0}$  and the  $q \times q$  Jacobian matrix  $\partial F / \partial \mathbf{x}'(\mathbf{x}_0, y_0)$  is invertible. Then there exist open neighbourhoods  $U_0 \subseteq U$  of  $\mathbf{x}_0$  and  $V_0 \subseteq V$  of  $y_0$  and a unique  $C^k$  function  $\varphi : V_0 \rightarrow U_0$  such that

$$F(\varphi(y), y) = \mathbf{0} \quad \text{for all } y \in V_0,$$

with  $\varphi(y_0) = \mathbf{x}_0$ .

### A.3 Proof of the main results

The proof of Lemma 2.1 and Corollary 2.1. is a direct application of the following two results.

**Lemma A.2.** *Let*

$$F(\boldsymbol{\theta}^*, \delta_0) := \mathbb{E}_{\delta_0}[\mathbf{s}_i^*(\boldsymbol{\theta}^*)], \quad (\text{A.12})$$

where  $\mathbb{E}_{\delta_0}$  denotes expectation under the true DGP with confounding strength  $\delta_0$ . Under conditions **R1–R7**, there exists a twice continuously differentiable function  $\delta_0 \mapsto \boldsymbol{\theta}^*(\delta_0)$  satisfying  $F(\boldsymbol{\theta}^*(\delta_0), \delta_0) = \mathbf{0}$  in a neighbourhood of  $\delta_0 = 0$ , with  $\boldsymbol{\theta}^*(0) = \boldsymbol{\theta}_0$  and

$$\boldsymbol{\theta}^*(\delta_0) = \boldsymbol{\theta}_0 + \boldsymbol{\Pi} \delta_0 + O(\delta_0^2), \quad (\text{A.13})$$

where

$$\boldsymbol{\Pi} := [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)]. \quad (\text{A.14})$$

Here  $\mathcal{I}^*(\boldsymbol{\theta}_0) := -\mathbb{E}_{\delta_0=0}[\mathbf{H}_i^*(\boldsymbol{\theta}_0)]$  is the short-model information matrix evaluated at the true parameter  $\boldsymbol{\theta}_0$ , where the short model is correctly specified. Thus  $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*(\delta_0)\| = O(\delta_0)$  as  $\delta_0 \rightarrow 0$ .

*Proof.* We work on  $\text{int}(\Xi)$ , which is open and contains  $(\boldsymbol{\theta}'_0, 0)'$  by **R1**. We verify the

conditions of Theorem A.1 with  $k = 2$ , applied to  $F(\boldsymbol{\theta}^*, \delta_0) = \mathbf{0}$  at the point  $(\boldsymbol{\theta}_0, 0)$ .

We note that  $F$  is twice continuously differentiable near  $(\boldsymbol{\theta}_0, 0)$  by Lemma A.1. Thus, we focus on the remaining two conditions. First, we show that  $F$  is 0 at  $(\boldsymbol{\theta}_0, 0)$ , so that this point is in the contour of  $F$ . This follows since, at  $\delta_0 = 0$  the short and long models coincide and the pseudo-true and true parameters agree, so  $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0$ . Applying (A.3) and score orthogonality (A.5):

$$F(\boldsymbol{\theta}_0, 0) = \mathbb{E}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0)] = \mathbb{E}_{\delta_0=0}[\mathbf{s}_{i,\theta}(\boldsymbol{\theta}_0, 0)] = \mathbf{0}. \quad (\text{A.15})$$

The last required conditions is invertibility of the the Jacobian with respect to  $\boldsymbol{\theta}^*$ . The Jacobian is

$$\left. \frac{\partial F}{\partial \boldsymbol{\theta}^{*\prime}} \right|_{(\boldsymbol{\theta}_0, 0)} = \mathbb{E}_{\delta_0=0}[\mathbf{H}_i^*(\boldsymbol{\theta}_0)] = -\mathcal{I}^*(\boldsymbol{\theta}_0), \quad (\text{A.16})$$

which is invertible by condition **R6**. Therefore, the conditions of Theorem A.1 with  $k = 2$  are all met, and there is a twice continuously differentiable function  $\delta_0 \mapsto \boldsymbol{\theta}^*(\delta_0)$  defined in a neighbourhood of  $\delta_0 = 0$  such that  $F(\boldsymbol{\theta}^*(\delta_0), \delta_0) = \mathbf{0}$  and  $\boldsymbol{\theta}^*(0) = \boldsymbol{\theta}_0$ .

We next derive the first derivative of  $\boldsymbol{\theta}^*(\cdot)$ . Totally differentiating  $F(\boldsymbol{\theta}^*(\delta_0), \delta_0) = \mathbf{0}$  with respect to  $\delta_0$  and rearranging gives

$$\frac{d\boldsymbol{\theta}^*}{d\delta_0} = - \left[ \frac{\partial F}{\partial \boldsymbol{\theta}^{*\prime}} \right]^{-1} \frac{\partial F}{\partial \delta_0}. \quad (\text{A.17})$$

Evaluating at  $\delta_0 = 0$  and using (A.16):

$$\left. \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right|_{\delta_0=0} = [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \left. \frac{\partial F}{\partial \delta_0} \right|_{(\boldsymbol{\theta}_0, 0)}. \quad (\text{A.18})$$

Since  $\delta_0$  enters  $F$  only through the distribution  $p(y, w; \delta_0) := p(y, w; \boldsymbol{\theta}_0, \delta_0)$  under which the expectation is taken, the derivative  $\partial F / \partial \delta_0$  produces the expectation of the integrand  $\mathbf{s}_i^*(\boldsymbol{\theta}^*)$  multiplied by the score of the true density with respect to  $\delta_0$ , namely  $s_{i,\delta}(\boldsymbol{\theta}_0, \delta_0) = \partial \log p(y, w; \delta_0) / \partial \delta_0$ . Specifically,

$$\frac{\partial F}{\partial \delta_0} = \frac{\partial}{\partial \delta_0} \int \mathbf{s}_i^*(\boldsymbol{\theta}^*) p(y, w; \delta_0) d\lambda(y, w) = \mathbb{E}_{\delta_0}[\mathbf{s}_i^*(\boldsymbol{\theta}^*) s_{i,\delta}(\boldsymbol{\theta}_0, \delta_0)], \quad (\text{A.19})$$

where the interchange of differentiation and integration was justified under the envelope assumptions.

At  $\delta_0 = 0$  the short model is correctly specified, so  $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0$ . Substituting into (A.19):

$$\left. \frac{\partial F}{\partial \delta_0} \right|_{(\boldsymbol{\theta}_0, 0)} = \mathbb{E}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0) s_{i,\delta}(\boldsymbol{\theta}_0, 0)]. \quad (\text{A.20})$$

By (A.7),  $\mathbb{E}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0)] = \mathbf{0}$ , and by (A.6),  $\mathbb{E}_0[s_{i,\delta}(\boldsymbol{\theta}_0, 0)] = 0$ . Therefore

$$\mathbb{E}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0) s_{i,\delta}(\boldsymbol{\theta}_0, 0)] = \text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)]. \quad (\text{A.21})$$

Substituting (A.21) into (A.18) gives

$$\left. \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right|_{\delta_0=0} = [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \text{Cov}_{\delta_0=0}[\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)] =: \boldsymbol{\Pi}. \quad (\text{A.22})$$

We now verify that the remainder is  $O(\delta_0^2)$ , for which  $\boldsymbol{\theta}^*(\cdot)$  needs to be in  $C^2$ . Totally differentiating

$$\frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \frac{d\boldsymbol{\theta}^*}{d\delta_0} + \frac{\partial F}{\partial \delta_0} = \mathbf{0} \quad (\text{A.23})$$

once more with respect to  $\delta_0$ , and applying the product rule to the first term:

$$\frac{d}{d\delta_0} \left( \frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right) = \frac{d}{d\delta_0} \left( \frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \right) \frac{d\boldsymbol{\theta}^*}{d\delta_0} + \frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \frac{d^2 \boldsymbol{\theta}^*}{d\delta_0^2}.$$

Because  $\partial F / \partial \boldsymbol{\theta}^{*'}$  depends on  $\delta_0$  both directly and through  $\boldsymbol{\theta}^*(\delta_0)$ , its total derivative is

$$\frac{d}{d\delta_0} \left( \frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \right) = \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'}^2} \frac{d\boldsymbol{\theta}^*}{d\delta_0} + \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'} \partial \delta_0}.$$

Similarly,

$$\frac{d}{d\delta_0} \left( \frac{\partial F}{\partial \delta_0} \right) = \frac{\partial^2 F}{\partial \delta_0 \partial \boldsymbol{\theta}^{*'}} \frac{d\boldsymbol{\theta}^*}{d\delta_0} + \frac{\partial^2 F}{\partial \delta_0^2}.$$

Since  $F \in C^2$ , the mixed partials coincide,  $\partial^2 F / (\partial \delta_0 \partial \boldsymbol{\theta}^{*'}) = \partial^2 F / (\partial \boldsymbol{\theta}^{*' \partial \delta_0})$ . Collecting

terms and setting the result equal to  $\mathbf{0}$ :

$$\frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \frac{d^2 \boldsymbol{\theta}^*}{d\delta_0^2} + \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'}2} \left[ \frac{d\boldsymbol{\theta}^*}{d\delta_0}, \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right] + 2 \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'} \partial \delta_0} \frac{d\boldsymbol{\theta}^*}{d\delta_0} + \frac{\partial^2 F}{\partial \delta_0^2} = \mathbf{0}. \quad (\text{A.24})$$

Since  $\partial F / \partial \boldsymbol{\theta}^{*'}$  is nonsingular in a sufficiently small neighbourhood of  $(\boldsymbol{\theta}_0, 0)$ :

$$\frac{d^2 \boldsymbol{\theta}^*}{d\delta_0^2} = - \left[ \frac{\partial F}{\partial \boldsymbol{\theta}^{*'}} \right]^{-1} \left\{ \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'}2} \left[ \frac{d\boldsymbol{\theta}^*}{d\delta_0}, \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right] + 2 \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'} \partial \delta_0} \frac{d\boldsymbol{\theta}^*}{d\delta_0} + \frac{\partial^2 F}{\partial \delta_0^2} \right\}. \quad (\text{A.25})$$

Evaluating at  $\delta_0 = 0$  and using  $d\boldsymbol{\theta}^*/d\delta_0|_{\delta_0=0} = \boldsymbol{\Pi}$ :

$$\left. \frac{d^2 \boldsymbol{\theta}^*}{d\delta_0^2} \right|_{\delta_0=0} = [\mathcal{I}^*(\boldsymbol{\theta}_0)]^{-1} \left\{ \left. \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'}2} \right|_{(\boldsymbol{\theta}_0, 0)} [\boldsymbol{\Pi}, \boldsymbol{\Pi}] + 2 \left. \frac{\partial^2 F}{\partial \boldsymbol{\theta}^{*'} \partial \delta_0} \right|_{(\boldsymbol{\theta}_0, 0)} \boldsymbol{\Pi} + \left. \frac{\partial^2 F}{\partial \delta_0^2} \right|_{(\boldsymbol{\theta}_0, 0)} \right\}. \quad (\text{A.26})$$

Each term on the right-hand side is finite: the first by **R5** ( $g_3$ ), the second by **R7** ( $g_{21}$ ), and the third by **R7** ( $g_{12}$ ). The  $C^2$  property implies that  $d^2 \boldsymbol{\theta}^*/d\delta_0^2$  is continuous and hence bounded in a sufficiently small neighbourhood of zero. Taylor's theorem with the Lagrange remainder then gives

$$\boldsymbol{\theta}^*(\delta_0) = \boldsymbol{\theta}^*(0) + \left. \frac{d\boldsymbol{\theta}^*}{d\delta_0} \right|_{\delta_0=0} \delta_0 + \frac{1}{2} \left. \frac{d^2 \boldsymbol{\theta}^*}{d\delta_0^2} \right|_{\tilde{\delta}_0} \delta_0^2 \quad (\text{A.27})$$

for some  $\tilde{\delta}_0$  between 0 and  $\delta_0$ . Since  $d^2 \boldsymbol{\theta}^*/d\delta_0^2$  is bounded locally, the last term is  $O(\delta_0^2)$ .

Using  $\boldsymbol{\theta}^*(0) = \boldsymbol{\theta}_0$  and  $d\boldsymbol{\theta}^*/d\delta_0|_{\delta_0=0} = \boldsymbol{\Pi}$  therefore yields

$$\boldsymbol{\theta}^*(\delta_0) = \boldsymbol{\theta}_0 + \boldsymbol{\Pi} \delta_0 + O(\delta_0^2),$$

establishing (A.13) and (A.14). Finally,  $\|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*(\delta_0)\| = O(\delta_0)$  as  $\delta_0 \rightarrow 0$ . ■

Note that, at  $\delta_0 = 0$ , the model is correctly specified and the information matrix equality gives  $\mathcal{I}^*(\boldsymbol{\theta}_0) = \mathbb{E}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0) \mathbf{s}_i^{*'}(\boldsymbol{\theta}_0)] = \text{Var}_0(\mathbf{s}_i^*(\boldsymbol{\theta}_0))$ . Therefore  $\boldsymbol{\Pi}$  is the population regression coefficient from regressing the score with respect to the confounding parameter  $\delta_0$  onto the short-model score evaluated at  $\boldsymbol{\theta}_0$ ; that is,

$$s_{i,\delta}(\boldsymbol{\theta}_0, 0) = \boldsymbol{\Pi}' \mathbf{s}_i^*(\boldsymbol{\theta}_0) + u_i, \quad \mathbb{E}_0[\mathbf{s}_i^*(\boldsymbol{\theta}_0) u_i] = \mathbf{0}. \quad (\text{A.28})$$

Hence  $\mathbf{\Pi}$  measures, to first order, how the confounder affects the short model parameter. We can take this interpretation further and explain the bias in terms of residuals from specific regressions similar to the linear OVB in Cinelli and Hazlett (2020).

**Corollary A.1** (Treatment coefficient bias). *Under the conditions of Lemma A.2, the bias in the treatment coefficient is*

$$\beta^* - \beta_0 = \tilde{\pi} \delta_0 + O(\delta_0^2), \quad (\text{A.29})$$

where

$$\tilde{\pi} = \frac{\tilde{\mathcal{C}}_{\beta\delta}^0}{\tilde{\mathcal{I}}_{\beta\beta}^0} = \frac{\text{Cov}_0(\tilde{s}_{\beta,i}, \tilde{s}_{\delta,i})}{\text{Var}_0(\tilde{s}_{\beta,i})}, \quad (\text{A.30})$$

where  $\tilde{s}_{\beta,i}$  and  $\tilde{s}_{\delta,i}$  are residualised scores.

*Proof.* Partition  $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \beta)'$  in the definition of  $\mathbf{\Pi}$  from Lemma A.2. Let

$$\mathcal{C}_{\boldsymbol{\theta}\delta}^0 := \text{Cov}_0(\mathbf{s}_i^*(\boldsymbol{\theta}_0), s_{i,\delta}(\boldsymbol{\theta}_0, 0)) = \begin{pmatrix} \mathcal{C}_{\alpha\delta}^0 \\ \mathcal{C}_{\beta\delta}^0 \end{pmatrix}, \quad (\text{A.31})$$

where  $\mathcal{C}_{\alpha\delta}^0 = \text{Cov}_0(\mathbf{s}_{\alpha,i}^*, s_{i,\delta}) \in \mathbb{R}^{(p+1) \times 1}$  and  $\mathcal{C}_{\beta\delta}^0 = \text{Cov}_0(s_{\beta,i}^*, s_{i,\delta}) \in \mathbb{R}^{1 \times 1}$ . The information matrix of the short model can be written as

$$\mathcal{I}^*(\boldsymbol{\theta}_0) = \begin{pmatrix} \mathcal{I}_{\alpha\alpha}^0 & \mathcal{I}_{\alpha\beta}^0 \\ \mathcal{I}_{\beta\alpha}^0 & \mathcal{I}_{\beta\beta}^0 \end{pmatrix}. \quad (\text{A.32})$$

Hence

$$\mathbf{\Pi} = \begin{pmatrix} \mathcal{I}_{\alpha\alpha}^0 & \mathcal{I}_{\alpha\beta}^0 \\ \mathcal{I}_{\beta\alpha}^0 & \mathcal{I}_{\beta\beta}^0 \end{pmatrix}^{-1} \begin{pmatrix} \mathcal{C}_{\alpha\delta}^0 \\ \mathcal{C}_{\beta\delta}^0 \end{pmatrix}. \quad (\text{A.33})$$

Write

$$\begin{pmatrix} \mathcal{I}_{\alpha\alpha}^0 & \mathcal{I}_{\alpha\beta}^0 \\ \mathcal{I}_{\beta\alpha}^0 & \mathcal{I}_{\beta\beta}^0 \end{pmatrix} \begin{pmatrix} \mathbf{\Pi}_\alpha \\ \mathbf{\Pi}_\beta \end{pmatrix} = \begin{pmatrix} \mathcal{C}_{\alpha\delta}^0 \\ \mathcal{C}_{\beta\delta}^0 \end{pmatrix}.$$

The first row gives

$$\mathbf{\Pi}_\alpha = (\mathcal{I}_{\alpha\alpha}^0)^{-1} (\mathcal{C}_{\alpha\delta}^0 - \mathcal{I}_{\alpha\beta}^0 \mathbf{\Pi}_\beta).$$

Substitute this into the second row:

$$\mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}(\mathcal{C}_{\alpha\delta}^0 - \mathcal{I}_{\alpha\beta}^0\Pi_\beta) + \mathcal{I}_{\beta\beta}^0\Pi_\beta = \mathcal{C}_{\beta\delta}^0.$$

Collecting the terms involving  $\Pi_\beta$ ,

$$(\mathcal{I}_{\beta\beta}^0 - \mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}\mathcal{I}_{\alpha\beta}^0)\Pi_\beta = \mathcal{C}_{\beta\delta}^0 - \mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}\mathcal{C}_{\alpha\delta}^0,$$

and hence

$$\Pi_\beta = (\mathcal{I}_{\beta\beta}^0 - \mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}\mathcal{I}_{\alpha\beta}^0)^{-1}(\mathcal{C}_{\beta\delta}^0 - \mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}\mathcal{C}_{\alpha\delta}^0). \quad (\text{A.34})$$

Define

$$\tilde{\mathcal{I}}_{\beta\beta}^0 := \mathcal{I}_{\beta\beta}^0 - \mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}\mathcal{I}_{\alpha\beta}^0, \quad (\text{A.35})$$

and

$$\tilde{\mathcal{C}}_{\beta\delta}^0 := \mathcal{C}_{\beta\delta}^0 - \mathcal{I}_{\beta\alpha}^0(\mathcal{I}_{\alpha\alpha}^0)^{-1}\mathcal{C}_{\alpha\delta}^0. \quad (\text{A.36})$$

It follows that

$$\Pi_\beta = \frac{\tilde{\mathcal{C}}_{\beta\delta}^0}{\tilde{\mathcal{I}}_{\beta\beta}^0}. \quad (\text{A.37})$$

At  $\boldsymbol{\xi}_0 = (\boldsymbol{\theta}'_0, 0)'$ , the short model is correctly specified. Hence the information identity gives

$$\mathcal{I}^*(\boldsymbol{\theta}_0) = \text{Var}_0(\mathbf{s}_i^*).$$

Thus, with

$$\mathbf{s}_i^* = \begin{pmatrix} \mathbf{s}_{\alpha,i}^* \\ s_{\beta,i}^* \end{pmatrix},$$

the blocks of  $\mathcal{I}^*(\boldsymbol{\theta}_0)$  satisfy

$$\mathcal{I}_{\alpha\alpha}^0 = \text{Var}_0(\mathbf{s}_{\alpha,i}^*), \quad \mathcal{I}_{\beta\alpha}^0 = \text{Cov}_0(s_{\beta,i}^*, \mathbf{s}_{\alpha,i}^*),$$

and

$$\mathcal{I}_{\alpha\beta}^0 = \text{Cov}_0(\mathbf{s}_{\alpha,i}^*, s_{\beta,i}^*).$$

Now consider the population projection of the treatment score  $s_{\beta,i}^*$  on the nuisance score  $\mathbf{s}_{\alpha,i}^*$ . Its residual is

$$\tilde{s}_{\beta,i} := s_{\beta,i}^* - \mathcal{I}_{\beta\alpha}^0 (\mathcal{I}_{\alpha\alpha}^0)^{-1} \mathbf{s}_{\alpha,i}^*. \quad (\text{A.38})$$

By the Schur-complement result in Lemma A.3, applied with

$$a_i = \mathbf{s}_{\alpha,i}^*, \quad b_i = s_{\beta,i}^*, \quad c_i = s_{i,\delta},$$

we obtain

$$\text{Var}_0(\tilde{s}_{\beta,i}) = \mathcal{I}_{\beta\beta}^0 - \mathcal{I}_{\beta\alpha}^0 (\mathcal{I}_{\alpha\alpha}^0)^{-1} \mathcal{I}_{\alpha\beta}^0 = \tilde{\mathcal{I}}_{\beta\beta}^0. \quad (\text{A.39})$$

The same lemma gives the corresponding residualisation of the  $\delta$  score,

$$\tilde{s}_{\delta,i} := s_{i,\delta} - \mathcal{C}_{\delta\alpha}^0 (\mathcal{I}_{\alpha\alpha}^0)^{-1} \mathbf{s}_{\alpha,i}^*, \quad (\text{A.40})$$

where

$$\mathcal{C}_{\delta\alpha}^0 := \text{Cov}_0(s_{i,\delta}, \mathbf{s}_{\alpha,i}^*) = (\mathcal{C}_{\alpha\delta}^0)'$$

The cross-residual covariance is therefore

$$\begin{aligned} \text{Cov}_0(\tilde{s}_{\beta,i}, \tilde{s}_{\delta,i}) &= \mathcal{C}_{\beta\delta}^0 - \mathcal{I}_{\beta\alpha}^0 (\mathcal{I}_{\alpha\alpha}^0)^{-1} \mathcal{C}_{\alpha\delta}^0 \\ &= \tilde{\mathcal{C}}_{\beta\delta}^0. \end{aligned} \quad (\text{A.41})$$

Consequently,

$$\Pi_\beta = \frac{\text{Cov}_0(\tilde{s}_{\beta,i}, \tilde{s}_{\delta,i})}{\text{Var}_0(\tilde{s}_{\beta,i})}. \quad (\text{A.42})$$

Combining this expression with  $\beta^* - \beta_0 = \Pi_\beta \delta_0 + O(\delta_0^2)$  from Lemma A.2 yields (A.29). ■

## A.4 GLMs with canonical link.

For GLMs with canonical link, the Hessian takes the form  $\mathbf{H}_i(\boldsymbol{\xi}) = -w_i(\boldsymbol{\xi}) W_i W_i'$ , where  $w_i(\boldsymbol{\xi}) = V(\mu_i(\boldsymbol{\xi})) > 0$  is the variance weight and  $W_i = (A_i', D_i, Z_i)' \in \mathbb{R}^{p+3}$  with  $A_i =$

$(1, \mathbf{X}'_i)' \in \mathbb{R}^{p+1}$ . At the pseudo-true parameter  $\boldsymbol{\xi}^* = (\boldsymbol{\theta}^{*'}, 0)'$ , write

$$w_i^* := V(\mu_i(\boldsymbol{\xi}^*)).$$

For the canonical link, the information matrix has the weighted second-moment representation

$$\mathcal{I}(\boldsymbol{\xi}^*) = \mathbb{E}[w_i^* W_i W_i']. \quad (\text{A.43})$$

Since  $\mathcal{I}(\boldsymbol{\xi}^*)$  is the second-moment matrix of  $\sqrt{w_i^*} W_i$ , the Schur-complement result in Lemma A.3 can be applied directly. Because the parameter subscripts  $(\boldsymbol{\alpha}, \beta, \delta)$  correspond to the regressors  $(A_i, D_i, Z_i)$ , the  $(\boldsymbol{\alpha}, \beta, \delta)$  partition of the information matrix is

$$\mathcal{I}(\boldsymbol{\xi}^*) = \begin{pmatrix} \mathcal{I}_{\alpha\alpha}^* & \mathcal{I}_{\alpha\beta}^* & \mathcal{I}_{\alpha\delta}^* \\ \mathcal{I}_{\beta\alpha}^* & \mathcal{I}_{\beta\beta}^* & \mathcal{I}_{\beta\delta}^* \\ \mathcal{I}_{\delta\alpha}^* & \mathcal{I}_{\delta\beta}^* & \mathcal{I}_{\delta\delta}^* \end{pmatrix}, \quad (\text{A.44})$$

where, for example,

$$\mathcal{I}_{\alpha\alpha}^* = \mathbb{E}[w_i^* A_i A_i'] \in \mathbb{R}^{(p+1) \times (p+1)}, \quad \mathcal{I}_{\beta\alpha}^* = \mathbb{E}[w_i^* D_i A_i'] \in \mathbb{R}^{1 \times (p+1)}, \quad \mathcal{I}_{\alpha\delta}^* = \mathbb{E}[w_i^* A_i Z_i] \in \mathbb{R}^{(p+1) \times 1}.$$

The Schur complement of the nuisance block  $\mathcal{I}_{\alpha\alpha}^*$  in the  $(\beta, \delta)$  submatrix is

$$\begin{pmatrix} \tilde{\mathcal{I}}_{\beta\beta}^* & \tilde{\mathcal{I}}_{\beta\delta}^* \\ \tilde{\mathcal{I}}_{\delta\beta}^* & \tilde{\mathcal{I}}_{\delta\delta}^* \end{pmatrix} = \begin{pmatrix} \mathcal{I}_{\beta\beta}^* & \mathcal{I}_{\beta\delta}^* \\ \mathcal{I}_{\delta\beta}^* & \mathcal{I}_{\delta\delta}^* \end{pmatrix} - \begin{pmatrix} \mathcal{I}_{\beta\alpha}^* \\ \mathcal{I}_{\delta\alpha}^* \end{pmatrix} (\mathcal{I}_{\alpha\alpha}^*)^{-1} \begin{pmatrix} \mathcal{I}_{\alpha\beta}^* & \mathcal{I}_{\alpha\delta}^* \end{pmatrix}. \quad (\text{A.45})$$

In particular,

$$\tilde{\mathcal{I}}_{\beta\beta}^* = \mathcal{I}_{\beta\beta}^* - \mathcal{I}_{\beta\alpha}^* (\mathcal{I}_{\alpha\alpha}^*)^{-1} \mathcal{I}_{\alpha\beta}^*, \quad (\text{A.46})$$

$$\tilde{\mathcal{I}}_{\beta\delta}^* = \mathcal{I}_{\beta\delta}^* - \mathcal{I}_{\beta\alpha}^* (\mathcal{I}_{\alpha\alpha}^*)^{-1} \mathcal{I}_{\alpha\delta}^*. \quad (\text{A.47})$$

To interpret these Schur-complement terms, apply Lemma A.3 with  $a_i = \sqrt{w_i^*} A_i$ ,  $b_i = \sqrt{w_i^*} D_i$ ,  $c_i = \sqrt{w_i^*} Z_i$ . The corresponding population projection residuals are

$$\tilde{D}_i^* = D_i - (\boldsymbol{\pi}_D^*)' A_i, \quad \boldsymbol{\pi}_D^* = (\mathcal{I}_{\alpha\alpha}^*)^{-1} \mathcal{I}_{\alpha\beta}^*, \quad (\text{A.48})$$

$$\tilde{Z}_i^* = Z_i - (\boldsymbol{\pi}_Z^*)' A_i, \quad \boldsymbol{\pi}_Z^* = (\mathcal{I}_{\alpha\alpha}^*)^{-1} \mathcal{I}_{\alpha\delta}^*, \quad (\text{A.49})$$

that is,  $\tilde{D}_i^*$  and  $\tilde{Z}_i^*$  are the residuals from  $w^*$ -weighted population projections of  $D_i$  and  $Z_i$  onto  $A_i$ . The Schur-complement identities in the lemma therefore give

$$\tilde{\mathcal{I}}_{\beta\beta}^* = \mathbb{E}[w_i^* \tilde{D}_i^{*2}], \quad (\text{A.50})$$

$$\tilde{\mathcal{I}}_{\beta\delta}^* = \mathbb{E}[w_i^* \tilde{D}_i^* \tilde{Z}_i^*]. \quad (\text{A.51})$$

Moreover, because  $A_i$  contains an intercept, the projection orthogonality conditions imply

$$\mathbb{E}[w_i^* \tilde{D}_i^*] = 0, \quad \mathbb{E}[w_i^* \tilde{Z}_i^*] = 0,$$

so the second moments in (A.50)–(A.51) are weighted variances and covariances. Defining the normalised weighted expectation

$$\mathbb{E}_w[X] := \frac{\mathbb{E}[w X]}{\mathbb{E}[w]},$$

we have

$$\text{Var}_{w^*}(\tilde{D}_i^*) = \frac{\tilde{\mathcal{I}}_{\beta\beta}^*}{\mathbb{E}[w_i^*]}, \quad \text{Cov}_{w^*}(\tilde{D}_i^*, \tilde{Z}_i^*) = \frac{\tilde{\mathcal{I}}_{\beta\delta}^*}{\mathbb{E}[w_i^*]}.$$

so

$$\tilde{\pi}(\delta_0) := \frac{\tilde{\mathcal{I}}_{\beta\delta}^*}{\tilde{\mathcal{I}}_{\beta\beta}^*} = \frac{\text{Cov}_{w^*}(\tilde{D}_i^*, \tilde{Z}_i^*)}{\text{Var}_{w^*}(\tilde{D}_i^*)}. \quad (\text{A.52})$$

Thus, for a canonical-link GLM, the pseudo-true coefficient sensitivity is exactly the population weighted regression slope of the residualised omitted regressor  $\tilde{Z}_i^*$  on the residualised treatment  $\tilde{D}_i^*$ , where the residualisation and weighting are both evaluated at the pseudo-true parameter.

Under **R2** and **R3**,  $w_i(\boldsymbol{\xi})$  is continuously differentiable in  $\boldsymbol{\xi}$  and the relevant weighted expectations are continuously differentiable by dominated convergence. Since  $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 + O(\delta_0)$  from Lemma A.2, the weighted moments defining the projection coefficients and residual moments satisfy

$$\tilde{\mathcal{I}}_{\beta\beta}^* = \tilde{\mathcal{I}}_{\beta\beta}^0 + O(\delta_0), \quad \tilde{\mathcal{I}}_{\beta\delta}^* = \tilde{\mathcal{I}}_{\beta\delta}^0 + O(\delta_0).$$

Provided  $\tilde{\mathcal{I}}_{\beta\beta}^0 > 0$  (ensured by **R6**), it follows that

$$\tilde{\pi}(\delta_0) = \tilde{\pi}(0) + O(\delta_0). \tag{A.53}$$

At  $\delta_0 = 0$  the model is correctly specified. Write  $w_i^0 := V(\mu_i(\boldsymbol{\xi}_0))$  and let  $\tilde{D}_i^0$  and  $\tilde{Z}_i^0$  denote the residuals from the  $w^0$ -weighted projections of  $D_i$  and  $Z_i$  onto  $A_i$ . Then

$$\tilde{\pi}(0) = \frac{\text{Cov}_{w^0}(\tilde{D}_i^0, \tilde{Z}_i^0)}{\text{Var}_{w^0}(\tilde{D}_i^0)}. \tag{A.54}$$

Combining (A.53) and (A.54) gives

$$\tilde{\pi}(\delta_0) = \frac{\text{Cov}_{w^0}(\tilde{D}_i^0, \tilde{Z}_i^0)}{\text{Var}_{w^0}(\tilde{D}_i^0)} + O(\delta_0). \tag{A.55}$$

Thus, in a neighbourhood of  $\delta_0 = 0$ , the first-order sensitivity coefficient is approximately the population regression slope from regressing  $\tilde{Z}_i^0$  on  $\tilde{D}_i^0$  using weights  $w_i^0$ . This is the nonlinear analogue of the OLS treatment-residual slope in the linear OVB formula of Cinelli and Hazlett (2020).

When  $h(\cdot)$  is the identity link and  $V(\mu_i) = \sigma^2$  is constant, the weights cancel, the weighted projections reduce to ordinary OLS projections, and (A.55) recovers the linear OVB formula exactly.

## A.5 Proof of Proposition 3.1

Let  $\tilde{D}^*$  (with  $\mathbb{E}[\tilde{D}^*] = 0$  and  $\text{Var}(\tilde{D}^*) = 1$ ),  $\tilde{Y}^*$  (with  $\mathbb{E}[\tilde{Y}^*] = 0$  and  $\text{Var}(\tilde{Y}^*) = 1$ ) and let  $\text{Cov}(\tilde{D}^*, \tilde{Y}^*) = r$  (and hence  $\text{Cor}(\tilde{D}^*, \tilde{Y}^*) = r$ ). The variable  $\varepsilon \sim N(0, 1)$  is independent of  $(\tilde{D}^*, \tilde{Y}^*)$ . Define the pseudo-confounder

$$Z = a \cdot \tilde{D}^* + b \cdot \tilde{Y}^* + c \cdot \varepsilon. \quad (\text{A.56})$$

We find  $a, b, c$  such that  $\text{Cor}(Z, \tilde{D}^*) = \rho_D$  and  $\text{Cor}(Z, \tilde{Y}^*) = \rho_Y$ . Noting that  $\mathbb{E}[Z] = 0$ , it follows that

$$\begin{aligned} \text{Cov}(Z, \tilde{D}^*) &= \text{Cov}(a\tilde{D}^* + b\tilde{Y}^* + c\varepsilon, \tilde{D}^*) \\ &= a \cdot \text{Cov}(\tilde{D}^*, \tilde{D}^*) + b \cdot \text{Cov}(\tilde{Y}^*, \tilde{D}^*) + c \cdot \text{Cov}(\varepsilon, \tilde{D}^*) \\ &= a + br \end{aligned} \quad (\text{A.57})$$

$$\begin{aligned} \text{Cov}(Z, \tilde{Y}^*) &= \text{Cov}(a\tilde{D}^* + b\tilde{Y}^* + c\varepsilon, \tilde{Y}^*) \\ &= ar + b \end{aligned} \quad (\text{A.58})$$

and

$$\begin{aligned} \text{Var}(Z) &= \text{Var}(a\tilde{D}^* + b\tilde{Y}^* + c\varepsilon) \\ &= a^2\text{Var}(\tilde{D}^*) + b^2\text{Var}(\tilde{Y}^*) + c^2\text{Var}(\varepsilon) \\ &\quad + 2ab\text{Cov}(\tilde{D}^*, \tilde{Y}^*) + 2ac\text{Cov}(\tilde{D}^*, \varepsilon) + 2bc\text{Cov}(\tilde{Y}^*, \varepsilon) \\ &= a^2 + b^2 + c^2 + 2abr. \end{aligned} \quad (\text{A.59})$$

Next, we can express correlations in terms of covariances. The correlation between  $Z$  and  $\tilde{D}^*$  is:

$$\text{Cor}(Z, \tilde{D}^*) = \frac{\text{Cov}(Z, \tilde{D}^*)}{\sqrt{\text{Var}(Z)} \cdot \sqrt{\text{Var}(\tilde{D}^*)}} = \frac{a + br}{\sqrt{a^2 + b^2 + c^2 + 2abr}} \quad (\text{A.60})$$

$$\text{Cor}(Z, \tilde{Y}^*) = \frac{ar + b}{\sqrt{a^2 + b^2 + c^2 + 2abr}}. \quad (\text{A.61})$$

Without loss of generality, we normalise  $Z$  to have unit variance:

$$a^2 + b^2 + c^2 + 2abr = 1 \quad \Rightarrow \quad c^2 = 1 - a^2 - b^2 - 2abr. \quad (\text{A.62})$$

With  $\text{Var}(Z) = 1$ , the correlations simplify to:

$$\text{Cor}(Z, \tilde{D}^*) = a + br \quad (\text{A.63})$$

$$\text{Cor}(Z, \tilde{Y}^*) = ar + b. \quad (\text{A.64})$$

Next, we solve for  $a$  and  $b$ . We require:

$$a + br = \rho_D \quad (\text{A.65})$$

$$ar + b = \rho_Y. \quad (\text{A.66})$$

This is a system of two linear equations in two unknowns ( $a$  and  $b$ ). Assuming  $r^2 \neq 1$  (i.e.,  $\tilde{D}^*$  and  $\tilde{Y}^*$  are not perfectly correlated)<sup>15</sup>:

$$b = \frac{\rho_Y - \rho_D r}{1 - r^2} \quad (\text{A.67})$$

$$a = \frac{\rho_D - \rho_Y r}{1 - r^2}. \quad (\text{A.68})$$

From the normalisation condition  $c^2 = 1 - a^2 - b^2 - 2abr$ , for a real solution we need  $c^2 \geq 0$ . Define  $u = \rho_D - \rho_Y r$  and  $v = \rho_Y - \rho_D r$  so that  $a = u/(1 - r^2)$  and  $b = v/(1 - r^2)$ .

Then,

$$a^2 + b^2 + 2abr = \frac{u^2 + v^2 + 2uvr}{(1 - r^2)^2}. \quad (\text{A.69})$$

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<sup>15</sup>To verify this result, note for example that

$$\begin{aligned} a + br &= \frac{\rho_D - \rho_Y r}{1 - r^2} + \frac{\rho_Y - \rho_D r}{1 - r^2} \cdot r = \frac{\rho_D - \rho_Y r + (\rho_Y - \rho_D r)r}{1 - r^2} \\ &= \frac{\rho_D - \rho_Y r + \rho_Y r - \rho_D r^2}{1 - r^2} = \frac{\rho_D(1 - r^2)}{1 - r^2} = \rho_D. \end{aligned}$$

Furthermore,

$$u^2 = \rho_D^2 - 2\rho_D\rho_Y r + \rho_Y^2 r^2 \quad (\text{A.70})$$

$$v^2 = \rho_Y^2 - 2\rho_D\rho_Y r + \rho_D^2 r^2 \quad (\text{A.71})$$

and

$$u^2 + v^2 = (\rho_D^2 + \rho_Y^2)(1 + r^2) - 4\rho_D\rho_Y r \quad (\text{A.72})$$

$$uv = \rho_D\rho_Y(1 + r^2) - r(\rho_D^2 + \rho_Y^2). \quad (\text{A.73})$$

It follows that

$$u^2 + v^2 + 2uvr = (1 - r^2)(\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Y r). \quad (\text{A.74})$$

Therefore:

$$a^2 + b^2 + 2abr = \frac{\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Y r}{1 - r^2}. \quad (\text{A.75})$$

The feasibility condition  $c^2 \geq 0$  becomes:

$$1 - \frac{\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Y r}{1 - r^2} = \frac{(1 - r^2) - (\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Y r)}{1 - r^2} \geq 0. \quad (\text{A.76})$$

Since  $1 - r^2 > 0$  (assuming  $|r| < 1$ ), this is equivalent to:

$$\rho_D^2 + \rho_Y^2 - 2\rho_D\rho_Y r \leq 1 - r^2. \quad (\text{A.77})$$

## A.6 Additional technical theorems and results.

The following is Lebesgue's Dominated Convergence Theorem as presented in Breuillard (2020), Theorem 7.13.

**Theorem A.2** (Lebesgue's Dominated Convergence Theorem). *Let  $f$  and  $(f_n)_{n \geq 1}$  be measurable functions on a measure space  $(X, \mathcal{A}, \mu)$ . Assume that there exists an integrable function  $g$  such that for all  $x \in X$ :*

$$(i) \quad |f_n(x)| \leq g(x) \text{ for all } n \geq 1,$$

$$(ii) \quad \lim_{n \rightarrow \infty} f_n(x) = f(x).$$

Then

$$\lim_{n \rightarrow \infty} \mu(f_n) = \mu(f),$$

and  $f$  is integrable.

The consequence of this result is that we can differentiate under the integral sign (Corollary 7.15 in Breuillard (2020))

**Corollary A.2** (Differentiation under the integral sign; Breuillard, 2020, Corollary 7.15).

*Let  $(X, \mathcal{A}, \mu)$  be a measure space and  $U \subset \mathbb{R}$  an open set. Let  $f : U \times X \rightarrow \mathbb{R}$  satisfy:*

$$(i) \quad \text{for every } t \in U, \text{ the function } f(t, \cdot) \text{ is } \mu\text{-integrable, i.e. } \int_X |f(t, x)| d\mu(x) < \infty;$$

$$(ii) \quad \text{for every } x \in X, \text{ the function } f(\cdot, x) \text{ is differentiable on } U;$$

$$(iii) \quad \text{there exists a } \mu\text{-integrable function } g : X \rightarrow [0, \infty) \text{ such that}$$

$$\left| \frac{\partial f}{\partial t}(t, x) \right| \leq g(x) \quad \text{for all } t \in U \text{ and all } x \in X.$$

Then for every  $t \in U$ , the function  $\frac{\partial f}{\partial t}(t, \cdot)$  is  $\mu$  integrable. Moreover, setting

$$F(t) := \int_X f(t, x) d\mu(x),$$

$F$  is differentiable on  $U$  and

$$F'(t) = \int_X \frac{\partial f}{\partial t}(t, x) d\mu(x).$$

Note that Corollary A.2 establishes that  $F$  is differentiable and gives an integral representation of  $F'$ , but does not by itself establish that  $F'$  is continuous, i.e. it does not give  $F \in C^1$ . Continuity of  $F'$  requires a separate argument, which we now provide together with the extension to higher orders.

**Lemma A.3** (Schur complements and population projection residuals). *Let  $a_i \in \mathbb{R}^p$ ,  $b_i \in \mathbb{R}^q$ , and  $c_i \in \mathbb{R}^r$  be random vectors with finite second moments. For any two such vectors  $u_i$  and  $v_i$ , write  $\mathcal{M}_{UV} := \mathbb{E}[u_i v_i']$ . Let  $\mathcal{M}$  denote the joint  $(p+q+r) \times (p+q+r)$  second-moment matrix of  $(a_i', b_i', c_i')'$ , partitioned as*

$$\mathcal{M} = \begin{pmatrix} \mathcal{M}_{AA} & \mathcal{M}_{AB} & \mathcal{M}_{AC} \\ \mathcal{M}_{BA} & \mathcal{M}_{BB} & \mathcal{M}_{BC} \\ \mathcal{M}_{CA} & \mathcal{M}_{CB} & \mathcal{M}_{CC} \end{pmatrix}, \quad (\text{A.78})$$

where  $\mathcal{M}_{AA} \in \mathbb{R}^{p \times p}$ ,  $\mathcal{M}_{AB} \in \mathbb{R}^{p \times q}$ ,  $\mathcal{M}_{AC} \in \mathbb{R}^{p \times r}$ , etc. Suppose that  $\mathcal{M}_{AA}$  is nonsingular. Define the population linear projection residuals

$$\tilde{b}_i = b_i - \Pi_{BA} a_i, \quad \Pi_{BA} = \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \in \mathbb{R}^{q \times p}, \quad (\text{A.79})$$

and, analogously,

$$\tilde{c}_i = c_i - \Pi_{CA} a_i, \quad \Pi_{CA} = \mathcal{M}_{CA} \mathcal{M}_{AA}^{-1} \in \mathbb{R}^{r \times p}. \quad (\text{A.80})$$

Then:

(i) The residuals are orthogonal to  $a_i$ :

$$\mathbb{E}[a_i \tilde{b}_i'] = \mathbf{0}_{p \times q}, \quad \mathbb{E}[a_i \tilde{c}_i'] = \mathbf{0}_{p \times r}. \quad (\text{A.81})$$

(ii) The cross second-moment matrix of the residuals is

$$\mathbb{E}[\tilde{b}_i \tilde{c}_i'] = \mathcal{M}_{BC} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AC} =: \mathcal{M}_{BC \cdot A} \in \mathbb{R}^{q \times r}. \quad (\text{A.82})$$

This is the  $(B, C)$  block of the Schur complement of  $\mathcal{M}_{AA}$  in  $\mathcal{M}$ :

$$\mathcal{M} / \mathcal{M}_{AA} = \begin{pmatrix} \mathcal{M}_{BB \cdot A} & \mathcal{M}_{BC \cdot A} \\ \mathcal{M}_{CB \cdot A} & \mathcal{M}_{CC \cdot A} \end{pmatrix} \in \mathbb{R}^{(q+r) \times (q+r)}. \quad (\text{A.83})$$

(iii) In particular, setting  $c_i = b_i$  gives the diagonal block

$$\mathbb{E}[\tilde{b}_i \tilde{b}_i'] = \mathcal{M}_{BB} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} = \mathcal{M}_{BB \cdot A} \in \mathbb{R}^{q \times q}. \quad (\text{A.84})$$

(iv) If either (a)  $\mathbb{E}[a_i] = \mathbb{E}[b_i] = \mathbb{E}[c_i] = \mathbf{0}$  or (b)  $a_i$  includes a constant component (intercept), then (A.82) is the covariance  $\text{Cov}(\tilde{b}_i, \tilde{c}_i)$  and (A.84) is the variance  $\text{Var}(\tilde{b}_i)$ .

*Proof.* To prove (i), note that by definition of  $\Pi_{BA}$ ,

$$\mathbb{E}[a_i \tilde{b}_i'] = \mathcal{M}_{AB} - \mathcal{M}_{AA} \Pi_{BA}' = \mathcal{M}_{AB} - \mathcal{M}_{AA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AB} = \mathbf{0}, \quad (\text{A.85})$$

and the argument for  $\tilde{c}_i$  is identical.

To prove (ii), partition  $\mathcal{M}$  into  $A$  and  $(B, C)$  blocks:

$$\mathcal{M} = \begin{pmatrix} \mathcal{M}_{AA} & \begin{pmatrix} \mathcal{M}_{AB} & \mathcal{M}_{AC} \end{pmatrix} \\ \begin{pmatrix} \mathcal{M}_{BA} \\ \mathcal{M}_{CA} \end{pmatrix} & \begin{pmatrix} \mathcal{M}_{BB} & \mathcal{M}_{BC} \\ \mathcal{M}_{CB} & \mathcal{M}_{CC} \end{pmatrix} \end{pmatrix}. \quad (\text{A.86})$$

The Schur complement of  $\mathcal{M}_{AA}$  in  $\mathcal{M}$  is then, by definition,

$$\begin{aligned}\mathcal{M}/\mathcal{M}_{AA} &= \begin{pmatrix} \mathcal{M}_{BB} & \mathcal{M}_{BC} \\ \mathcal{M}_{CB} & \mathcal{M}_{CC} \end{pmatrix} - \begin{pmatrix} \mathcal{M}_{BA} \\ \mathcal{M}_{CA} \end{pmatrix} \mathcal{M}_{AA}^{-1} \begin{pmatrix} \mathcal{M}_{AB} & \mathcal{M}_{AC} \end{pmatrix} \\ &= \begin{pmatrix} \mathcal{M}_{BB \cdot A} & \mathcal{M}_{BC \cdot A} \\ \mathcal{M}_{CB \cdot A} & \mathcal{M}_{CC \cdot A} \end{pmatrix}.\end{aligned}$$

Then, expand

$$\begin{aligned}\mathbb{E}[\tilde{b}_i \tilde{c}_i'] &= \mathbb{E}[(b_i - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} a_i) (c_i - \mathcal{M}_{CA} \mathcal{M}_{AA}^{-1} a_i)'] \\ &= \mathcal{M}_{BC} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AC} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AC} + \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AC} \\ &= \mathcal{M}_{BC} - \mathcal{M}_{BA} \mathcal{M}_{AA}^{-1} \mathcal{M}_{AC},\end{aligned}\tag{A.87}$$

proving (A.82). Setting  $c_i = b_i$  yields (A.84). To prove (iv), note that when  $a_i$  includes a constant, the corresponding row of  $a_i b_i'$  is  $b_i'$  itself, and the orthogonality condition  $\mathbb{E}[a_i \tilde{b}_i'] = \mathbf{0}$  implies  $\mathbb{E}[\tilde{b}_i] = \mathbf{0}$  (and likewise for  $\tilde{c}_i$ ), so  $\mathbb{E}[\tilde{b}_i \tilde{c}_i'] = \text{Cov}(\tilde{b}_i, \tilde{c}_i)$ . ■

## A.7 Application to Multinomial Logit models

This subsection considers the application of the synthetic method to a multinomial logit model, as these are models of great empirical interest in economics, psychology, social sciences and other behavioural disciplines.

In a multinomial logit with  $J > 2$  categories, each category has its own set of structural parameters and its own residual. In particular, with  $J$  categories and baseline category  $j = 0$ , we have

$$P(Y_i = j \mid D_i, \mathbf{X}_i, Z_i) = \frac{\exp(\alpha_{0,j} + \beta_j D_i + \boldsymbol{\gamma}_j' \mathbf{X}_i + \delta_j Z_i)}{1 + \sum_{k \neq 0} \exp(\alpha_{0,k} + \beta_k D_i + \boldsymbol{\gamma}_k' \mathbf{X}_i + \delta_k Z_i)}, \quad j \neq 0, \tag{A.88}$$

where the subscript  $j$  indexes the outcome category. Each category  $j$  has its own coeffi-

cient on the omitted confounder  $\delta_j$ . The AME of  $D$  on  $P(Y_i = j)$  is

$$\widehat{\text{AME}}_j = \frac{1}{n} \sum_{i=1}^n \hat{P}(Y_i = j) [\beta_j - \sum_{k \neq 0} \beta_k \hat{P}(Y_i = k)], \quad (\text{A.89})$$

and each category produces a separate AME that must be assessed independently.

Define the residual for category  $j$  as

$$\tilde{Y}_{j,i} = \mathbf{1}(Y_i = j) - \hat{P}(Y_i = j \mid D_i, \mathbf{X}_i), \quad j = 1, \dots, J. \quad (\text{A.90})$$

If one computes  $Z$  using only category  $j$ 's residual, then  $Z$  achieves the target  $\rho_Y$  with  $\tilde{Y}_j$  by construction, but not with the remaining  $\tilde{Y}_k$  for  $k \neq j$ . The consequence is that including such a  $Z$  in the multinomial logit would simultaneously shift the AMEs of all  $J$  categories, but in uncontrolled directions for  $k \neq j$ , and we cannot interpret the resulting sensitivity surface for category  $j$  as the effect of a confounder with a specific  $(\rho_D, \rho_Y)$  relationship to category  $j$  alone.

The solution proposed here is to construct a separate synthetic confounder  $Z_j$  for each outcome category  $j$ . Each  $Z_j$  is built using category  $j$ 's own residual  $\tilde{Y}_j$  in the loading formula, so that its target correlations  $(\rho_D, \rho_Y)$  are defined with respect to outcome  $j$ . The multinomial logit is then re-estimated with  $Z_j$  included and the AME for category  $j$  is recorded. This is repeated independently for every category.

Formally, for each category  $j$ :

1. Compute the standardised residuals from the short model:  $\tilde{Y}_{j,i} = \mathbf{1}(Y_i = j) - \hat{P}(Y_i = j \mid D_i, \mathbf{X}_i)$ , and standardise:  $\tilde{Y}_j^* = (\tilde{Y}_j - \bar{\tilde{Y}}_j) / \hat{\sigma}_{\tilde{Y}_j}$ .
2. Compute the treatment–outcome residual correlation  $r_j = \text{Cor}(\tilde{D}^*, \tilde{Y}_j^*)$ .
3. Given target  $(\rho_D, \rho_Y)$  and  $r_j$ , solve for the loadings  $(a_j, b_j, c_j)$  via:

$$a_j = \frac{\rho_D - \rho_Y r_j}{1 - r_j^2}, \quad b_j = \frac{\rho_Y - \rho_D r_j}{1 - r_j^2}, \quad c_j = \sqrt{1 - a_j^2 - b_j^2 - 2a_j b_j r_j}.$$

4. Construct  $Z_j = a_j \tilde{D}^* + b_j \tilde{Y}_j^* + c_j \varepsilon$ , where  $\varepsilon \sim N(0, 1)$  independent of everything

else and where  $\text{Cor}(Z_j, \tilde{D}^*) = \rho_D$  and  $\text{Cor}(Z_j, \tilde{Y}_j^*) = \rho_Y$  by construction.

5. Re-estimate the multinomial logit including  $Z_j$  as an additional covariate and record  $\widehat{\text{AME}}_j$ .

Steps 1–5 are repeated for each  $(\rho_D, \rho_Y)$  combination in the grid and for each category  $j \in \{1, \dots, J\}$ .

Unlike with single outcome models, there will be an outcome-specific feasibility region requiring  $c_j^2 \geq 0$ , which expands to:

$$1 - a_j^2 - b_j^2 - 2a_j b_j r_j \geq 0. \quad (\text{A.91})$$

with the feasible set of  $(\rho_D, \rho_Y)$  being the interior of the ellipse:

$$\rho_D^2 + \rho_Y^2 - 2r_j \rho_D \rho_Y \leq 1 - r_j^2, \quad (\text{A.92})$$

The bisection for the explain-away threshold  $\rho_D^*$  and sign-change threshold  $\rho_D^+$  is executed separately for each outcome category using that category's own  $r_j$ , residual  $\tilde{Y}_j^*$ , and confounder  $Z_j$ .

Note that the  $\varepsilon$  draw used to construct  $Z_j$  is held fixed across all  $(\rho_D, \rho_Y)$  grid points within each category, but is shared across categories within the same grid call. This ensures that the sensitivity surface for each category is smooth (the fixed  $\varepsilon$  eliminates Monte Carlo noise).

The following simulation study illustrates the use of the synthetic confounder method on a three-outcome multinomial logit, showing how the analysis can reveal different robustness properties across outcome categories. As in Section 5 of the paper, the DGP generates  $n = 20,000$  observations with the same treatment assignment and controls as before ( $X_j \sim N(0, 1)$ ,  $D = \mathbf{1}(0.3X_1 + u > 0)$ ). The confounder is  $Z = 0.3\tilde{D}^* + \sqrt{1 - 0.09}\eta$ ,

giving  $\rho_D^{\text{true}} = 0.3$ . Specifically, the outcome has three categories with baseline  $j = 1$ :

$$P(Y_i = 2 \mid \cdot) \propto \exp(-0.5 + 0.5 D_i + 0.2 X_{1i} + 0.3 X_{2i} + 0.4 Z_i), \quad (\text{A.93})$$

$$P(Y_i = 3 \mid \cdot) \propto \exp(-1.0 + 0.7 D_i + 0.1 X_{1i} + 0.4 X_{2i} + 0.5 Z_i). \quad (\text{A.94})$$

Since the multinomial probabilities must sum to one, treatment increases the probability of categories 2 and 3 at the expense of category 1, yielding a negative AME for category 1 and positive AMEs for categories 2 and 3. More precisely, the shot model AMEs are -0.183 (0.007), 0.072 (0.007) and 0.111 (0.006) for outcomes 1 to 3 respectively.

Tables 9–11 report the adjusted AMEs across the  $(\rho_D, \rho_Y)$  grid for each outcome category. As expected, the  $\rho_D = 0$  column is flat (a confounder uncorrelated with treatment cannot bias the AME), the  $\rho_Y = 0$  row is flat (a variable uncorrelated with the outcome residual is not a confounder). Elsewhere the AME varies smoothly as both partial correlations increase.

Because the AME for outcome 1 is negative, a confounder positively associated with treatment and negatively associated with the outcome residual would inflate the magnitude of the negative effect. That is,  $\rho_D > 0$  combined with  $\rho_Y < 0$  will overstate the true AME. Conversely, for outcomes 2 and 3 (positive AMEs), the AME is reduced (toward zero) as we move toward positive  $\rho_D$  and positive  $\rho_Y$ .

Table 12 reports the explain-away and sign-change thresholds for all three outcomes. As expected, there are substantive differences in robustness across categories, driven by the magnitude of the short-model AME.

Outcome 1 has the largest AME in absolute value ( $-0.183$ ,  $t \approx 28$ ) and is accordingly the most robust. For  $\rho_Y \in [0.0, 0.3]$  (confounders whose association with the outcome residual would attenuate the negative effect), no feasible  $\rho_D$  can make the AME insignificant or reverse its sign. Even at  $\rho_Y = -0.5$  (the most damaging direction for a negative AME), the explain-away threshold is  $\rho_D^* = 0.456$ , requiring a confounder that explains over 20% of the residual treatment variance.

Note that the thresholds at  $\rho_Y < 0$  are positive ( $\rho_D^* > 0$ ), while those at  $\rho_Y > 0$  are

**Table 9:** Multinomial logit, outcome 1 (short-model AME =  $-0.183$ ): adjusted AME.

| $\rho_D$ | $\rho_Y$ |        |        |        |        |        |        |        |        |        |        |
|----------|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|          | -0.5     | -0.4   | -0.3   | -0.2   | -0.1   | 0.0    | 0.1    | 0.2    | 0.3    | 0.4    | 0.5    |
| -0.50    | -0.322   | -0.307 | -0.285 | -0.255 | -0.220 | -0.181 | -0.141 | -0.102 | -0.063 | -0.024 | 0.014  |
| -0.40    | -0.300   | -0.288 | -0.270 | -0.245 | -0.215 | -0.181 | -0.147 | -0.114 | -0.080 | -0.046 | -0.013 |
| -0.30    | -0.275   | -0.266 | -0.252 | -0.233 | -0.208 | -0.182 | -0.155 | -0.128 | -0.101 | -0.074 | -0.047 |
| -0.20    | -0.247   | -0.241 | -0.231 | -0.218 | -0.201 | -0.182 | -0.163 | -0.144 | -0.126 | -0.107 | -0.087 |
| -0.10    | -0.216   | -0.213 | -0.208 | -0.201 | -0.192 | -0.183 | -0.173 | -0.163 | -0.153 | -0.143 | -0.133 |
| 0.00     | -0.184   | -0.184 | -0.184 | -0.184 | -0.184 | -0.183 | -0.183 | -0.183 | -0.183 | -0.183 | -0.181 |
| 0.10     | -0.151   | -0.154 | -0.159 | -0.166 | -0.174 | -0.184 | -0.194 | -0.203 | -0.213 | -0.222 | -0.230 |
| 0.20     | -0.117   | -0.123 | -0.133 | -0.147 | -0.165 | -0.185 | -0.205 | -0.225 | -0.245 | -0.264 | -0.281 |
| 0.30     | -0.081   | -0.090 | -0.104 | -0.126 | -0.154 | -0.185 | -0.217 | -0.249 | -0.281 | -0.312 | -0.339 |
| 0.40     | -0.039   | -0.050 | -0.070 | -0.101 | -0.140 | -0.186 | -0.232 | -0.279 | -0.325 | -0.369 | -0.408 |
| 0.50     | 0.010    | -0.003 | -0.028 | -0.069 | -0.123 | -0.187 | -0.252 | -0.317 | -0.381 | -0.442 | -0.494 |

**Table 10:** Multinomial logit, outcome 2 (short-model AME =  $0.072$ ): adjusted AME.

| $\rho_D$ | $\rho_Y$ |        |        |        |       |       |       |        |        |        |        |
|----------|----------|--------|--------|--------|-------|-------|-------|--------|--------|--------|--------|
|          | -0.5     | -0.4   | -0.3   | -0.2   | -0.1  | 0.0   | 0.1   | 0.2    | 0.3    | 0.4    | 0.5    |
| -0.50    | -0.064   | -0.050 | -0.029 | -0.001 | 0.033 | 0.070 | 0.109 | 0.147  | 0.185  | 0.223  | 0.259  |
| -0.40    | -0.043   | -0.031 | -0.014 | 0.009  | 0.038 | 0.071 | 0.104 | 0.137  | 0.170  | 0.202  | 0.232  |
| -0.30    | -0.018   | -0.009 | 0.004  | 0.022  | 0.045 | 0.071 | 0.097 | 0.124  | 0.150  | 0.176  | 0.200  |
| -0.20    | 0.011    | 0.016  | 0.025  | 0.038  | 0.053 | 0.071 | 0.090 | 0.109  | 0.127  | 0.145  | 0.161  |
| -0.10    | 0.041    | 0.044  | 0.048  | 0.054  | 0.063 | 0.072 | 0.081 | 0.091  | 0.101  | 0.110  | 0.117  |
| 0.00     | 0.073    | 0.073  | 0.072  | 0.072  | 0.072 | 0.072 | 0.072 | 0.073  | 0.073  | 0.073  | 0.071  |
| 0.10     | 0.105    | 0.102  | 0.097  | 0.090  | 0.082 | 0.073 | 0.063 | 0.054  | 0.044  | 0.035  | 0.025  |
| 0.20     | 0.138    | 0.132  | 0.123  | 0.110  | 0.092 | 0.073 | 0.053 | 0.034  | 0.014  | -0.006 | -0.025 |
| 0.30     | 0.174    | 0.166  | 0.152  | 0.131  | 0.104 | 0.074 | 0.042 | 0.011  | -0.020 | -0.051 | -0.081 |
| 0.40     | 0.214    | 0.204  | 0.186  | 0.157  | 0.118 | 0.074 | 0.029 | -0.016 | -0.061 | -0.106 | -0.147 |
| 0.50     | 0.262    | 0.251  | 0.228  | 0.189  | 0.136 | 0.075 | 0.011 | -0.052 | -0.115 | -0.176 | -0.231 |

**Table 11:** Multinomial logit, outcome 3 (short-model AME =  $0.111$ ): adjusted AME.

| $\rho_D$ | $\rho_Y$ |       |       |       |       |       |       |       |        |        |        |
|----------|----------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|
|          | -0.5     | -0.4  | -0.3  | -0.2  | -0.1  | 0.0   | 0.1   | 0.2   | 0.3    | 0.4    | 0.5    |
| -0.50    | -0.010   | 0.003 | 0.021 | 0.047 | 0.077 | 0.110 | 0.144 | 0.178 | 0.212  | 0.244  | 0.273  |
| -0.40    | 0.008    | 0.019 | 0.035 | 0.056 | 0.082 | 0.110 | 0.140 | 0.169 | 0.198  | 0.225  | 0.250  |
| -0.30    | 0.030    | 0.038 | 0.050 | 0.067 | 0.088 | 0.111 | 0.134 | 0.157 | 0.180  | 0.202  | 0.221  |
| -0.20    | 0.054    | 0.060 | 0.069 | 0.080 | 0.095 | 0.111 | 0.127 | 0.144 | 0.160  | 0.175  | 0.187  |
| -0.10    | 0.081    | 0.084 | 0.089 | 0.095 | 0.103 | 0.111 | 0.119 | 0.128 | 0.136  | 0.144  | 0.149  |
| 0.00     | 0.108    | 0.109 | 0.110 | 0.111 | 0.111 | 0.111 | 0.111 | 0.111 | 0.112  | 0.111  | 0.109  |
| 0.10     | 0.136    | 0.135 | 0.132 | 0.127 | 0.120 | 0.111 | 0.103 | 0.095 | 0.086  | 0.078  | 0.069  |
| 0.20     | 0.165    | 0.162 | 0.155 | 0.144 | 0.129 | 0.112 | 0.094 | 0.077 | 0.059  | 0.043  | 0.026  |
| 0.30     | 0.195    | 0.191 | 0.180 | 0.163 | 0.139 | 0.112 | 0.084 | 0.056 | 0.029  | 0.003  | -0.021 |
| 0.40     | 0.230    | 0.224 | 0.210 | 0.185 | 0.151 | 0.112 | 0.072 | 0.032 | -0.007 | -0.045 | -0.077 |
| 0.50     | 0.270    | 0.264 | 0.246 | 0.214 | 0.167 | 0.112 | 0.056 | 0.000 | -0.055 | -0.105 | -0.147 |

**Table 12:** Multinomial logit: fragility thresholds by outcome category.

|          | Outcome 1<br>(AME = -0.183) |            | Outcome 2<br>(AME = 0.072) |            | Outcome 3<br>(AME = 0.111) |            |
|----------|-----------------------------|------------|----------------------------|------------|----------------------------|------------|
| $\rho_Y$ | $\rho_D^*$                  | $\rho_D^+$ | $\rho_D^*$                 | $\rho_D^+$ | $\rho_D^*$                 | $\rho_D^+$ |
| -0.50    | 0.456                       | 0.481      | -0.195                     | -0.237     | -0.387                     | -0.443     |
| -0.40    | 0.480                       | 0.505      | -0.214                     | -0.262     | -0.442                     | -0.518     |
| -0.30    | 0.528                       | 0.554      | -0.256                     | -0.321     | -0.586                     | -0.763     |
| -0.20    | 0.614                       | 0.641      | -0.367                     | -0.489     | robust                     | robust     |
| -0.10    | 0.762                       | 0.785      | robust                     | robust     | robust                     | robust     |
| 0.00     | robust                      | robust     | 0.992                      | robust     | 0.996                      | robust     |
| 0.10     | robust                      | robust     | 0.483                      | 0.551      | 0.656                      | 0.693      |
| 0.20     | robust                      | robust     | 0.292                      | 0.344      | 0.463                      | 0.500      |
| 0.30     | robust                      | robust     | 0.205                      | 0.243      | 0.351                      | 0.382      |
| 0.40     | -0.561                      | -0.664     | 0.158                      | 0.187      | 0.281                      | 0.307      |
| 0.50     | -0.402                      | -0.445     | 0.128                      | 0.151      | 0.236                      | 0.257      |

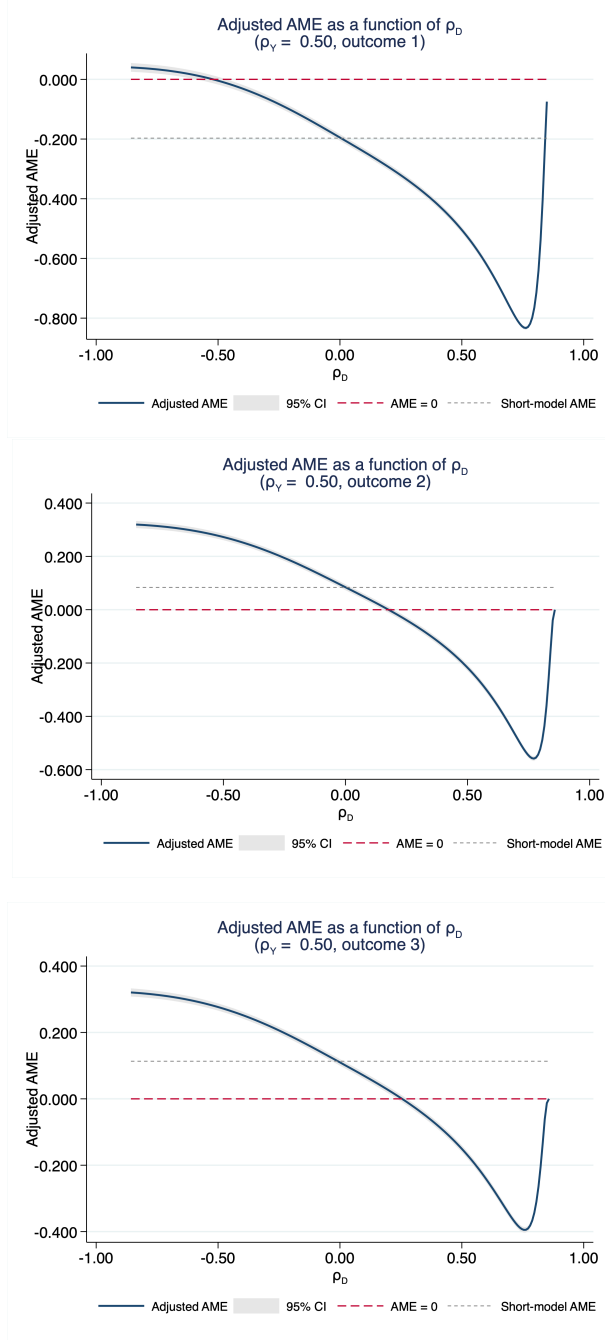
Notes:  $\rho_D^*$  is the explain-away threshold;  $\rho_D^+$  is the sign-change threshold. “robust” indicates no threshold exists within the feasible range. Each outcome has its own synthetic confounder  $Z_j$ , feasibility ellipse, and residual correlation  $r_j$ .

negative ( $\rho_D^* < 0$ ). Again, this reflects the fact that the short-model AME is negative. A confounder that inflates a negative effect must be positively correlated with treatment ( $\rho_D > 0$ ) and negatively correlated with the outcome residual ( $\rho_Y < 0$ ), so that treated units have lower outcome residuals, making  $D$  appear more harmful than it is. Conversely, to attenuate a negative effect toward zero requires a confounder working in the opposite direction ( $\rho_D < 0$ ,  $\rho_Y > 0$ ).

Outcome 2 has the smallest AME (0.072,  $t \approx 11$ ) and is substantially more fragile. At  $\rho_Y = 0.5$ , the explain-away threshold is only  $\rho_D^* = 0.128$ . That is, a confounder explaining just 1.6% of the residual treatment variance would suffice to render this effect insignificant.

Outcome 3 occupies an intermediate position (AME = 0.111,  $t \approx 19$ ). Its thresholds are approximately 50–70% larger than those of outcome 2 at every  $\rho_Y$ . At  $\rho_Y = 0.5$ , the explain-away threshold is  $\rho_D^* = 0.236$ , and the sign-change threshold is  $\rho_D^+ = 0.257$ . At  $\rho_Y = 0.0$ , the threshold is near the feasibility boundary ( $\rho_D^* = 0.996$ ), and for  $\rho_Y \in [-0.2, -0.1]$  the result is robust.

Figure 1 displays the sensitivity profile for each outcome category at  $\rho_Y = 0.5$ , show-



**Figure 1:** Sensitivity profiles for the multinomial logit at  $\rho_Y = 0.5$ . Each panel shows the adjusted AME as a function of  $\rho_D$  across the feasible interval, with 95% confidence bands (shaded), the zero line (dashed), and the short-model AME (dotted). The zero crossing identifies the sign-change threshold  $\rho_D^+$ .

ing the adjusted AME as a function of  $\rho_D$  across the full feasible interval. All three profiles share a smooth, monotonically decreasing curve that eventually plunges near the feasibility boundary where  $c^2 \rightarrow 0$ . The main difference is where the curve crosses zero (the sign-change threshold) and in how far from  $\rho_D = 0$  this occurs. For outcome 1, the curve crosses zero at  $\rho_D \approx -0.5187$ , indicating that a confounder would need to explain roughly 19% of the residual treatment variance to reverse the sign of this effect. For outcome 3, the zero crossing occurs at  $\rho_D \approx 0.2529$  (approximately 7% of residual treatment variance). For outcome 2 (AME = 0.084), the crossing occurs at  $\rho_D \approx 0.15$  (just 2% of residual treatment variance) confirming the fragility identified in Table 12. The near-boundary behaviour visible at the right edge of each panel (the sharp dip and partial recovery) reflects the point where the synthetic confounder’s independent noise  $\varepsilon$  vanishes ( $c \rightarrow 0$ ) and the augmented model becomes collinear; these values lie outside the substantively interpretable region and are excluded from the threshold computation by the scan margin.

Across all three outcomes, the sign-change threshold  $|\rho_D^+|$  exceeds the explain-away threshold  $|\rho_D^*|$  whenever both exist, as expected. The gap between the two thresholds is smallest for outcome 2, where the weak AME means insignificance and sign reversal occur at nearly the same confounding strength, and largest for outcome 1, where the strong AME creates a wider band between losing significance and actually flipping sign.

Table 13 reports the benchmark analysis for each outcome category.

**Table 13:** Multinomial logit: benchmark analysis by outcome.

| Outcome | Variable | $\hat{\rho}_D$ | $\hat{\rho}_Y$ | Adj. AME | SE    | $t$   | Bias   |
|---------|----------|----------------|----------------|----------|-------|-------|--------|
| 1       | $X_1$    | 0.225          | -0.052         | -0.173   | 0.007 | -25.8 | -0.011 |
|         | $X_2$    | 0.005          | -0.149         | -0.183   | 0.006 | -28.5 | 0.000  |
| 2       | $X_1$    | 0.225          | 0.062          | 0.059    | 0.007 | 8.8   | 0.013  |
|         | $X_2$    | 0.005          | 0.063          | 0.072    | 0.007 | 11.0  | 0.000  |
| 3       | $X_1$    | 0.225          | -0.010         | 0.114    | 0.006 | 18.7  | -0.002 |
|         | $X_2$    | 0.005          | 0.103          | 0.111    | 0.006 | 18.9  | 0.000  |

Notes: bias =  $\widehat{\text{AME}}_{\text{short}} - \widehat{\text{AME}}_{\text{adj}}$ .  $\hat{\rho}_D$  is identical across outcomes (it depends only on the treatment equation), while  $\hat{\rho}_Y$  is category-specific.

$\hat{\rho}_D$  is identical across outcomes for each benchmark variable, since it depends only on

the relationship between the control and the treatment residual, which is invariant to the outcome category. The  $\hat{\rho}_Y$ , depend on the category because each outcome has its own residual.

For  $X_2$ ,  $\hat{\rho}_D \approx 0$  across all outcomes, so  $X_2$  provides no useful benchmarking information regardless of its association with the outcome residual. This is expected since  $X_2$  does not enter the treatment equation.

For  $X_1$ ,  $\hat{\rho}_D = 0.225$  is moderate. Its impact varies across outcomes. For outcome 1, the bias of  $-0.011$  is small relative to the AME of  $-0.183$ , and the adjusted  $t$ -statistic of  $-25.8$  leaves the result highly significant. The negative bias (i.e., the adjustment makes the AME slightly more negative) is consistent with  $X_1$  having  $\hat{\rho}_Y = -0.052$  for this outcome. That is, a confounder with  $\rho_D > 0$  and  $\rho_Y < 0$  inflates the magnitude of a negative AME, so adjusting for it reduces the magnitude slightly.

For outcome 2, the  $X_1$  benchmark produces a bias of  $0.013$ , which is 18% of the short-model AME. Comparing  $\hat{\rho}_D = 0.225$  against the explain-away thresholds in Table 12 at  $\hat{\rho}_Y \approx 0.06$ : interpolating between  $\rho_Y = 0.0$  ( $\rho_D^* = 0.992$ ) and  $\rho_Y = 0.1$  ( $\rho_D^* = 0.483$ ),  $X_1$ 's treatment-confounding falls short of the threshold, so the result survives confounding at  $X_1$ 's strength. However, a confounder only moderately stronger than  $X_1$  on the outcome dimension could overturn outcome 2's significance.

For outcome 3,  $X_1$  has a near-zero outcome-confounding ( $\hat{\rho}_Y = -0.010$ ), producing negligible bias ( $-0.002$ ). This illustrates how the same observed control can be a meaningful benchmark for one outcome category and uninformative for another.

## B Supporting Results

**Table 14:** Partial Correlations of Age-10 Cognitive Skills (PC1) with Covariates

|                 | Partial | Semipartial | Partial            | $p$ -value |
|-----------------|---------|-------------|--------------------|------------|
| Variable        | corr.   | corr.       | corr. <sup>2</sup> |            |
| <i>Outcomes</i> |         |             |                    |            |

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Table 14 continued from previous page

| Variable                                           | Partial<br>corr. | Semipartial<br>corr. | Partial<br>corr. <sup>2</sup> | p-value |
|----------------------------------------------------|------------------|----------------------|-------------------------------|---------|
| Employed full-time                                 | 0.0427           | 0.0343               | 0.0018                        | 0.038   |
| Manager or supervisor                              | 0.0628           | 0.0505               | 0.0039                        | 0.002   |
| <i>Non-cognitive skills</i>                        |                  |                      |                               |         |
| Perseverance (age 10, SD)                          | 0.2807           | 0.2346               | 0.0788                        | 0.000   |
| <i>Birth and perinatal characteristics (Set 1)</i> |                  |                      |                               |         |
| Mother's age at delivery                           | −0.0003          | −0.0003              | 0.0000                        | 0.987   |
| Mother's age at delivery <sup>2</sup>              | 0.0120           | 0.0096               | 0.0001                        | 0.560   |
| Region 1                                           | 0.0230           | 0.0184               | 0.0005                        | 0.266   |
| Region 2                                           | 0.0219           | 0.0176               | 0.0005                        | 0.289   |
| Region 3                                           | −0.0006          | −0.0005              | 0.0000                        | 0.975   |
| Region 4                                           |                  | (dropped)            |                               |         |
| Region 5                                           | 0.0172           | 0.0138               | 0.0003                        | 0.404   |
| Region 6                                           | −0.0109          | −0.0088              | 0.0001                        | 0.596   |
| Region 7                                           | 0.0050           | 0.0040               | 0.0000                        | 0.810   |
| Region 8                                           | 0.0158           | 0.0126               | 0.0002                        | 0.445   |
| Region 9                                           | −0.0233          | −0.0187              | 0.0005                        | 0.260   |
| Region 10                                          | 0.0221           | 0.0178               | 0.0005                        | 0.283   |
| Mother smoked during pregnancy                     | −0.0289          | −0.0232              | 0.0008                        | 0.162   |
| Abnormal gestation                                 | −0.0310          | −0.0249              | 0.0010                        | 0.133   |
| Female                                             | −0.0575          | −0.0462              | 0.0033                        | 0.005   |
| Birth weight                                       | 0.0431           | 0.0346               | 0.0019                        | 0.037   |
| <i>Age-5 family background (Set 2)</i>             |                  |                      |                               |         |
| Mother stayed in education                         | 0.0719           | 0.0579               | 0.0052                        | 0.001   |
| Father stayed in education                         | 0.0561           | 0.0451               | 0.0031                        | 0.007   |

Continued on next page

Table 14 continued from previous page

| Variable                                               | Partial<br>corr. | Semipartial<br>corr. | Partial<br>corr. <sup>2</sup> | p-value |
|--------------------------------------------------------|------------------|----------------------|-------------------------------|---------|
| Parity                                                 | −0.0684          | −0.0550              | 0.0047                        | 0.001   |
| Previous stillbirth                                    | 0.0346           | 0.0278               | 0.0012                        | 0.093   |
| Previous preterm birth                                 | −0.0075          | −0.0060              | 0.0001                        | 0.717   |
| Previous stillbirth (high)                             | 0.0100           | 0.0080               | 0.0001                        | 0.627   |
| Previous miscarriages                                  | −0.0022          | −0.0018              | 0.0000                        | 0.914   |
| Breastfed                                              | 0.0416           | 0.0334               | 0.0017                        | 0.044   |
| <i>Age-5 health (Set 3)</i>                            |                  |                      |                               |         |
| Inpatient stay                                         | 0.0080           | 0.0064               | 0.0001                        | 0.699   |
| Hearing or sight problems                              | −0.0219          | −0.0176              | 0.0005                        | 0.288   |
| Child health rating                                    | 0.0194           | 0.0156               | 0.0004                        | 0.347   |
| <i>Parental education (Set 3 cont.)</i>                |                  |                      |                               |         |
| Mother: non-level qualification                        | −0.0572          | −0.0460              | 0.0033                        | 0.006   |
| Mother: secondary qualification                        | −0.0330          | −0.0265              | 0.0011                        | 0.110   |
| Mother: degree                                         | 0.0412           | 0.0330               | 0.0017                        | 0.046   |
| Father: non-level qualification                        | −0.0260          | −0.0209              | 0.0007                        | 0.208   |
| Father: secondary qualification                        | 0.0072           | 0.0058               | 0.0001                        | 0.727   |
| Father: degree                                         | 0.0351           | 0.0282               | 0.0012                        | 0.089   |
| <i>Maternal employment and household (Set 3 cont.)</i> |                  |                      |                               |         |
| Mother employed full-time (age 5)                      | −0.0120          | −0.0096              | 0.0001                        | 0.561   |
| Mother employed part-time (age 5)                      | −0.0020          | −0.0016              | 0.0000                        | 0.922   |
| Non-white                                              | −0.0409          | −0.0329              | 0.0017                        | 0.047   |
| No siblings at age 5                                   | −0.0019          | −0.0015              | 0.0000                        | 0.927   |
| <i>Age-10 health and home environment (Set 4)</i>      |                  |                      |                               |         |
| Child health (age 10)                                  | −0.0103          | −0.0082              | 0.0001                        | 0.619   |

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Table 14 continued from previous page

| Variable                                 | Partial<br>corr. | Semipartial<br>corr. | Partial<br>corr. <sup>2</sup> | p-value |
|------------------------------------------|------------------|----------------------|-------------------------------|---------|
| Inpatient stay (age 10)                  | 0.0157           | 0.0126               | 0.0002                        | 0.447   |
| Father unemployed (age 10)               | −0.0131          | −0.0105              | 0.0002                        | 0.527   |
| Father stay-at-home (age 10)             | −0.0206          | −0.0165              | 0.0004                        | 0.319   |
| Mother unemployed (age 10)               | −0.0194          | −0.0156              | 0.0004                        | 0.347   |
| Mother stay-at-home (age 10)             | −0.0035          | −0.0028              | 0.0000                        | 0.867   |
| Benefit receipt                          | 0.0276           | 0.0221               | 0.0008                        | 0.181   |
| Income band 1                            | 0.0317           | 0.0255               | 0.0010                        | 0.124   |
| Income band 2                            |                  | (dropped)            |                               |         |
| Income band 3                            | 0.0454           | 0.0364               | 0.0021                        | 0.028   |
| Income band 4                            | 0.0426           | 0.0342               | 0.0018                        | 0.039   |
| Income band 5                            | 0.0288           | 0.0231               | 0.0008                        | 0.162   |
| Income band 6                            | 0.0557           | 0.0448               | 0.0031                        | 0.007   |
| Income band 7                            | 0.0480           | 0.0386               | 0.0023                        | 0.020   |
| <i>Accommodation type (Set 5)</i>        |                  |                      |                               |         |
| Accommodation type 1                     | −0.0220          | −0.0177              | 0.0005                        | 0.286   |
| Accommodation type 2                     | −0.0202          | −0.0162              | 0.0004                        | 0.328   |
| Accommodation type 3                     | −0.0268          | −0.0215              | 0.0007                        | 0.194   |
| Accommodation type 4                     | −0.0250          | −0.0201              | 0.0006                        | 0.225   |
| Accommodation type 5                     |                  | (dropped)            |                               |         |
| Accommodation type 6                     | −0.0227          | −0.0182              | 0.0005                        | 0.271   |
| Accommodation type 7                     | −0.0161          | −0.0129              | 0.0003                        | 0.437   |
| Number of rooms                          | 0.0241           | 0.0194               | 0.0006                        | 0.242   |
| <i>Parental job satisfaction (Set 6)</i> |                  |                      |                               |         |
| Mother's job satisfaction                | −0.0155          | −0.0124              | 0.0002                        | 0.454   |

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Table 14 continued from previous page

| Variable                                                         | Partial<br>corr. | Semipartial<br>corr. | Partial<br>corr. <sup>2</sup> | p-value |
|------------------------------------------------------------------|------------------|----------------------|-------------------------------|---------|
| Father's job satisfaction                                        | −0.0210          | −0.0169              | 0.0004                        | 0.308   |
| <i>Medical and sensory (Set 6 cont.)</i>                         |                  |                      |                               |         |
| Eyesight problems                                                | −0.0002          | −0.0002              | 0.0000                        | 0.992   |
| Hearing problems                                                 | −0.0396          | −0.0318              | 0.0016                        | 0.055   |
| Speech problems                                                  | 0.0257           | 0.0207               | 0.0007                        | 0.212   |
| <i>Physical development</i>                                      |                  |                      |                               |         |
| Height (adjusted)                                                | 0.1095           | 0.0884               | 0.0120                        | 0.000   |
| Weight (adjusted)                                                | 0.1082           | 0.0874               | 0.0117                        | 0.000   |
| <i>Teacher questionnaire and classroom behaviour PCs (Set 7)</i> |                  |                      |                               |         |
| Perserverance (age 10)                                           | −0.0329          | −0.0264              | 0.0011                        | 0.111   |
| Behaviour PC B                                                   | 0.1536           | 0.1247               | 0.0236                        | 0.000   |
| Behaviour PC D                                                   | −0.0423          | −0.0340              | 0.0018                        | 0.040   |
| Behaviour PC E                                                   | −0.0028          | −0.0022              | 0.0000                        | 0.893   |
| Behaviour PC F                                                   | −0.0436          | −0.0350              | 0.0019                        | 0.034   |
| Behaviour PC G                                                   | −0.0045          | −0.0036              | 0.0000                        | 0.829   |
| Behaviour PC H                                                   | −0.0036          | −0.0029              | 0.0000                        | 0.860   |

Notes:  $N = 2,422$ . Partial and semipartial correlations from `pcorr` in Stata, conditioning on the full covariate set. PC1 is the first principal component of five age-10 cognitive test scores.  $p$ -values are two-tailed. Variables marked (dropped) were collinear with other regressors.